Disaster-Exposed Peers and Unequal Academic Loss: Evidence from the Wenchuan Earthquake

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Abstract

This study examines intragroup peer effects among students affected by the 2008 Great Sichuan Earthquake in China using original survey data linked to administrative test scores. Exploiting quasi-random classroom allocation following school transition after the earthquake, I show that a higher classroom share of school-collapsed peers significantly lowers the academic performance of less-exposed students. These losses are largest for low-rank students and for female students. The mechanism evidence is more consistent with deterioration in classroom relationships and interpersonal support than with broad conduct disruption, rank competition, reduced study effort, or teacher-side explanations. By examining spillovers among disaster-affected students themselves, rather than spillovers from affected newcomers onto unaffected incumbents, this paper complements mainly intergroup studies of post-disaster peer effects and extends them by incorporating psychosocial and relational evidence alongside academic outcomes.

Keywords: Peer effects, Disasters, Human Capital Development

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1 Introduction

Natural disasters are increasingly common and pose major risks to long-run development by disrupting the accumulation of human capital, particularly in low- and middle-income countries with fragile educational systems (UNESCO, 2023). While the direct effects of disasters on affected students, through school closures, infrastructure damage, and displacement, are well studied (Morrill and Westall, 2023; Andrabi, Daniels, and Das, 2021; Tian, Gong, and Zhai, 2022), far less is known about how these shocks reverberate through the social network in education. In practice, learning is held in classrooms, peer groups, and networks. When disaster-affected students are grouped together during school recovery, these peer interactions may amplify academic deficiency.

This study investigates the role of peer effects among disaster affected students during the schooling resumption phase following the 2008 Great Sichuan Earthquake in China. Existing literature assesses how exposure to natural disasters affects the academic performance of students who do not directly experience these events, typically by studying non-affected incumbent students in schools that receive evacuees from disaster-affected areas (Imberman, Kugler, and Sacerdote, 2012; Figlio and Özek, 2019). However, students with common disaster experiences may exhibit different peer effects. Disaster exposure commonly induces post-traumatic responses in children and adolescents, including PTSD symptoms, elevated anxiety and depression, and disrupted attention and engagement (Lai et al., 2018; Deuchert and Felfe, 2015; Tian, Gong, and Zhai, 2022). When students who share such experiences are placed in the same classroom, the psychological distress that each carries may be reinforced through repeated peer interactions. This reinforcement, in turn, can degrade the classroom environment, transmitting the initial disaster shock into further academic loss through a mechanism that is not visible when affected and unaffected students are not together in the same classroom.

This study examines a unique post-disaster schooling context in Wenchuan County, Sichuan, China. The central question is whether students with more severe earthquake experiences further affect their classmates' academic performance. The setting is particularly well suited to this question because students affected by the earthquake resumed schooling within a short time frame, were reassigned into new classrooms during recovery, and continued learning in a disaster-affected student population rather than being dispersed into unaffected local classrooms. The analysis focuses on academic outcomes in Chinese and mathematics during post-disaster recovery.

The analysis uses a unique combination of survey and administrative data. The *Kin Mirai Kadai Kaiketsu Jigyo* project surveyed earthquake-affected counties in December 2009. This study focuses on Wenchuan County, where the project covered all elementary, middle, and high schools. The survey collected information on the schools students attended during the earthquake and on the schools in which they were enrolled at the time of the survey, about one and a half years after the earthquake. At the student level, it records psychosocial and behavioral outcomes, background characteristics, and earthquake-related experiences. These survey data are linked to administrative test scores from the 2008 post-earthquake examination, taken before students entered their new classrooms, and from the 2010 end-of-year examination, collected roughly half a year after the survey. Earthquake exposure is proxied by the extent of school damage, given that most students were inside school buildings when the earthquake struck. Because the educational bureau required instruction to resume within two months regardless of the level of school damage, school damage provides a measure of direct earthquake experience that is not mechanically confounded by variation in school-closure duration.

I identify how earthquake exposure affects other peers by leveraging three natural settings: (i) the level of earthquake exposure, indicated by the collapse of old schools, was unanticipated and uncorrelated with student characteristics;[1] (ii) students had a natural transition from elementary/middle schools (old schools) to middle/high schools (new schools) after the earthquake, and (iii) students were assigned to classrooms in their new schools by a mechanical rank-rotation rule that dispersed students of different achievement levels across classrooms rather than sorting them on unobservable characteristics, making classroom composition as-good-as-random within each school-grade cell, or within each school-grade-track cell in the tracked school. The first setting was induced by the unprecedented earthquake features. The second setting ensured a diverse mixture of classmates for each student, with varying average levels of peer earthquake exposure. This variation in peer exposure is generated by the mixing of old-school cohorts at school transition. Students advancing from primary to middle school, or from middle to high school, enrolled in new schools that drew students from multiple old schools with different earthquake damage. The classroom-assignment rule then redistributed this mixed cohort across classrooms within each new school-grade cell, so that classrooms within the same school-grade differ in the share of school-collapsed peers. To effectively utilize this natural transition, the study examines only students in the 7th, 8th, 10th, and 11th grades, as these students had progressed to a higher level of post-earthquake schooling at the time of the survey. Lastly, the third setting ensures that the key variable of interest, peer earthquake exposure, is not confounded by unobserved factors. This exogenous formation of peer groups facilitates the isolation of the peer effect from potential unobserved clustering biases.

[1]The building damage is a frequently used index to measure the damage caused by a disaster. For instance, Deuchert and Felfe (2015) uses house damage to proxy for typhoon damage, whereas Raeburn (2023) uses school damage to represent destruction caused by a hurricane.

To estimate the effect of peer exposure on academic performance, I use a fixed-effects model with school-grade fixed effects and an academic-track indicator,[2] so identification comes from differences in the share of school-collapsed classmates across classrooms within the same school-grade cell, or within the same school-grade-track cell in the tracked school. Under the quasi-random classroom-allocation rule, this within-cell variation is plausibly exogenous conditional on the pre-enrollment test score.

In terms of academic performance, I find that students assigned to classrooms with a higher share of school-collapsed classmates perform worse in Chinese, with weaker negative patterns for mathematics and total scores. Students whose own old school collapsed also perform worse on academic outcomes. These results survive a range of robustness checks that rule out alternative exposure definitions, concerns about class size and closely related classroom-resource differences such as pupil-teacher ratio, selective sample construction, missing 2010 outcomes, accidental classroom composition, and omitted-variable bias as the main explanation.

Beyond academic outcomes, the analysis also examines psychosocial, behavioral, and well-being outcomes, including depression, behavioral and peer difficulties, self-esteem, family environment, happiness, health, and study motivation. Own school collapse is associated with worse subjective wellbeing, especially lower self-reported health. Peer exposure mainly worsens peer relationships and subjective wellbeing: a higher share of school-collapsed classmates markedly increases peer problems and reduces self-reported happiness and health, but does not significantly affect broader psychosocial measures such as depression and self-esteem.

These academic losses are unevenly distributed across students. The direct effect of own school collapse on test scores and overall rank is markedly larger for students at the bottom of the baseline rank distribution, while the peer effect of being assigned more school-collapsed classmates is markedly larger for low-rank and female students. Overall, the pattern points to greater sensitivity to adverse peer exposure among low-rank and female students.

I next examine which mechanisms could transmit the peer-exposure effect. Prior literature suggests several possible channels, including classroom disruption, peer and student-teacher relationships, rank competition, teacher pedagogical practice, and changes in student effort. The evidence points most strongly to a classroom environment channel operating through both peer relationships and student-teacher relationships. A higher share of school-collapsed classmates worsens peer problems, low-rank students report less peer and teacher comfort, and female students show a stronger peer-relationship response to same-gender exposure. These patterns line up with the heterogeneity results: academic losses are largest for low-rank students, and peer-exposure losses are also larger for female students. By contrast, the evidence does not support classroom disruption, rank competition, sorting of teachers across classrooms,

[2] Only one school in the sample operates an academic-tracking system, which divides students into a higher-performing and a lower-performing track; the quasi-random classroom assignment applies within each track.

changes in teacher pedagogical practice, or changes in student effort as the main explanation for the academic spillover.

To the best of my knowledge, this study is the first to examine intragroup peer effects on academic scores among students affected by disasters. When students who shared a traumatic experience are placed in the same classroom, their proximity may deepen post-traumatic responses. The intergroup literature, including Imberman, Kugler, and Sacerdote (2012) and Figlio and Özek (2019), focuses by design on the incumbent peers of disaster-displaced newcomers rather than on the affected students themselves, so the peer-effect consequences for the affected students' own academic and psychosocial trajectories remain unexamined. More fundamentally, when disaster-affected students are dispersed among unaffected classmates, the two groups do not share a common traumatic experience, and the channel through which co-exposure operates only becomes active when affected students are grouped together in the same classroom, a configuration that intergroup designs cannot generate. The recent work of Tincani (2025) does study disaster-affected students, but her design leverages the disaster-induced reshuffling of within-classroom academic rankings to identify rank-competition spillovers, and her outcomes center on academic performance, with less attention to psychosocial and relational dimensions. My setting pairs quasi-random classroom assignment with survey measures of mental health, behavior, and interpersonal relationships, identifying the interpersonal-relationship channel as the primary mechanism through which co-exposed peers reinforce each other's academic loss. Within this setting, I show that a higher share of school-collapsed classmates lowers academic performance and worsens peer relationships and subjective wellbeing among disaster-exposed students themselves.

More broadly, this paper makes three contributions to the literature. First, it extends existing work on post-disaster peer effects in education. The dominant strand of this literature examines how disaster- or conflict-displaced newcomers affect the academic outcomes of unaffected incumbent students in receiving schools: Imberman, Kugler, and Sacerdote (2012) on Hurricane Katrina evacuees in Houston and Louisiana, Figlio and Özek (2019) on Haitian-earthquake migrants in Florida, Özek (2023) on Hurricane Maria displacees, Morales (2022) on refugee resettlement, and Padilla-Romo and Peluffo (2023) on violence-displaced peers. The spillovers these studies document work largely through the administrative and compositional asymmetry between affected newcomers and unaffected incumbents: compensatory funding flows to schools that receive displaced students (Morales 2022), teacher quality is reallocated from advanced to remedial classrooms (Özek 2023), and the average ability composition of the classroom shifts (Imberman, Kugler, and Sacerdote 2012). These channels exist because affected and unaffected students share a classroom but are differentially treated, and they act in part by crowding out attention and resources from incumbents. In my setting, every classmate has experienced school collapse, so the school-level policy responses that underlie these channels, such as compensatory school funding tied to displaced enrollments and the reallocation of teacher resources across classrooms, do not apply. Instead, changes in the classroom environment arising from shared disaster experience plausibly play a more central role in shaping the intragroup peer effect.

Second, this paper contributes to research on disasters and social relations. One strand estimates how disasters affect affected students' academic or enrollment outcomes (Sacerdote 2012; Di Pietro 2018; Brück, Di Maio, and Miaari 2019; Tian, Gong, and Zhai 2022; Andrabi, Daniels, and Das 2021; Deuchert and Felfe 2015; Gallagher, Billings, and Ricketts 2023; Raeburn 2025; Callen 2015). In this literature, disaster exposure is often intertwined with variation in school closure duration or schooling interruption. The mandatory uninterrupted-schooling policy in my setting helps isolate the school-collapse experience from this source of variation. A separate strand examines social capital in resettlement and recovery, focusing on self-selected community ties among adults (Higuchi et al. 2024; Cao, Xu, and Zhang 2022; Shoji and Murata 2021; Bazzi et al. 2019; Hikichi et al. 2017; Johar et al. 2022). Relative to this literature, I study students whose post-disaster peer networks are shaped by school policy rather than self-selection, and I examine psychosocial and relational outcomes alongside academic performance.

This study also contributes to the broader literature on the effects of peer composition on academic performance in classrooms. A large body of work links a wide range of peer characteristics to classmates' academic and behavioral outcomes, including exposure to pollution, only-child status, troubled family background, race, maternal education, ethnicity, peer ability and tracking, social-interaction patterns and comparative advantage, classroom rank, peer gender composition, and peer behavior (Gazze, Persico, and Spirovska, 2024; Cai, Fan, and Yuan, 2022; Zhao and Zhao, 2021; Carrell, Hoekstra, and Kuka, 2018; d'Este and Einiö, 2021; Nguyen, 2024; de Gendre et al., 2024; Duflo, Dupas, and Kremer, 2011; Patacchini, Rainone, and Zenou, 2017; Cicala, Fryer, and Spenkuch, 2018; Murphy and Weinhardt, 2020; Rosenzweig and Xu, 2023; Carneiro et al., 2025; Yu, 2020; Getik and Meier, 2022; Barwick et al., 2026). The standard research design exploits plausibly exogenous variation in peer composition across classes, grades, or cohorts, with proposed mechanisms working through classroom climate, social interactions, and information transmission. My analysis aligns methodologically with this literature and extends it by using shared disaster experience as the source of variation in peer composition.

This paper proceeds as follows. Section 2 provides the context for the direct impact of the Great Sichuan Earthquake on students, school resumption, and examination arrangements throughout the recovery period. Section 3 describes the survey and variables. Section 4 introduces the empirical strategy and identification issues. Section 5 presents the empirical results, with Section 5.1 covering academic outcomes and Section 5.2 covering psychosocial and wellbeing outcomes. Section 6 examines heterogeneity in these effects. Section 7 discusses potential mechanisms and alternative explanations. Section 8 concludes.

2 Background

2.1 Intensity of the Great Sichuan Earthquake

Measuring at 8.0 Ms, the Great Sichuan Earthquake hit the Wenchuan County of the Sichuan province in China at 2:28 pm on May 12, 2008. The earthquake resulted in 69,197 deaths, 374,176 injuries, and 18,222 missing people (Ministry of Civil Affairs, 2008).

It is noteworthy that students were exposed to a greater extent of damage owing to weak school buildings. Throughout Sichuan Province, over 7,000 schoolrooms collapsed after the earthquake, leading to the deaths of over 5,000 students. The number of crumbled school buildings is disproportionately high owing to shoddy construction.[3] In some cases, schools collapsed, whereas nearby buildings were only slightly damaged (Ministry of Civil Affairs, 2008). When the earthquake occurred, the majority of students were on a noon break inside the school buildings, where they unpreparedly experienced heavily shaking schoolhouses. Surviving students in Sichuan Province suffered from being buried beneath the rubble of a collapsed building or getting injured.

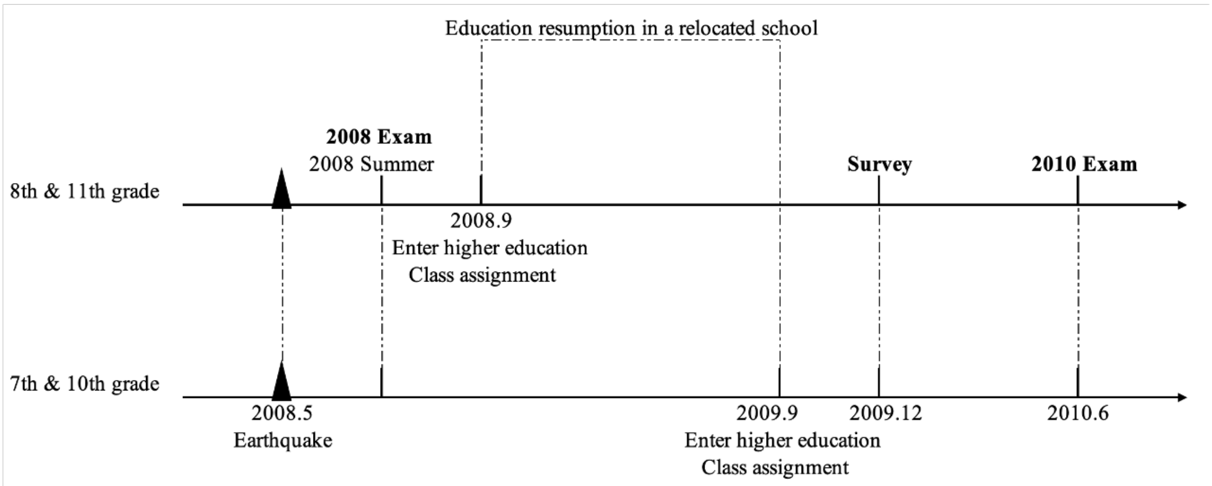
2.2 School resumption, exams, and class assignment

Under the instructions of the Ministry of Education, all schools resumed schooling before the beginning of the 2008 autumn semester. To restart instruction as quickly as possible, students first studied in temporary classrooms or were relocated to host schools, and then returned to newly built permanent school buildings by 2009. In the most heavily affected county, Wenchuan, many students studied in host schools outside the county during the first post-earthquake year. Under this arrangement, students relocated as intact classes and continued studying with their original classmates from Wenchuan, rather than being integrated into local classrooms in the host schools.

The relocation itself therefore did not change students' peer group. Students who remained at the same school level continued to study with the same classmates even if instruction temporarily took place in a host school outside the county. The relevant change in classmates occurred only for students who advanced to a new school level after the earthquake. Students moving from primary to middle school or from middle to high school enrolled in a new school and were assigned to new classrooms at entry. This enrollment-time assignment mixed students

[3]There has been extensive media coverage on the destruction of school buildings following the earthquake. See Wong (2008).

Figure 1: Timeline



from different old schools across classrooms, generating within-school-grade variation in the share of earthquake-affected peers. Although official county-level statistics do not separately report permanent schooling-related outmigration from Wenchuan, several institutional features make large-scale educational outmigration unlikely.[4]

Figure 1 shows the sequence of the earthquake, end-of-year exams, and class assignments. The earthquake struck in May 2008, approximately one month before the scheduled end-of-year exams, which were postponed and taken during the summer of that year. By September 2008, all students had resumed school.

The timing of grade advancement is particularly important in this context. Students in the 8th and 11th grades at the time of the survey in December 2009 had advanced to a higher school level by the beginning of the 2008–2009 academic year. Similarly, students in the 7th and 10th grades at the time of the survey moved up a grade before the 2009–2010 academic year. Consequently, for these four grades, the scores from the 2008 end-of-academic year exam, conducted immediately after the earthquake, served as the pre-enrollment scores for the new school. The scores from the 2010 academic year, when the survey was conducted, were considered the outcomes of interest in this study. Therefore, hereafter, the “pre-enrollment score” and “2008 score” both refer to the scores obtained immediately following the earthquake, while the “2010 score” is used to assess academic performance influenced by their peers.

In Wenchuan County, schools’ approach to class assignments after the earthquake followed their usual procedure: newly enrolled students were sorted into different classes at the start

[4] China’s compulsory education system emphasized nearby exam-free enrollment, typically linked to the student’s hukou location, and compulsory education was tuition- and fee-free by this period (State Council of the People’s Republic of China (2006); Sichuan Provincial Department of Education (2006)). Wenchuan was also a poor, predominantly ethnic-minority county (The Central People’s Government of the People’s Republic of China (2008)), making a switch into non-local private schools less plausible. Official reports further indicate that post-earthquake student movement mainly took the form of temporary off-site school resumption.

of their first academic year in the new school. Classroom assignment was conducted only once, implying that classmate composition remained stable from the first assignment until graduation. Regardless of schools and grades, the common study life between students and their classmates in 2010 lasted at least one year before the 2010 end-year exam. The empirical analysis therefore compares classrooms within the same school-grade group that differ in the share of students whose old schools collapsed because newly enrolled students were quasi-randomly allocated across classrooms at school entry. The details of the assignment rule and balance evidence are discussed in Section 4.2.

3 Survey & Data

3.1 Survey on Student Development in the Post-disaster Recovery

This research employs a subset of data from a unique survey conducted in December 2009, in conjunction with administrative test scores. The survey covered earthquake-affected counties in Sichuan and focused on the cognitive and noncognitive outcomes of affected children.[5] This study focuses on schools in Wenchuan County, which experienced the highest earthquake severity.

The research team collected students' samples using the following procedures. All elementary, middle, and high schools in Wenchuan County constituted the Wenchuan school sample. Within each school, 50% of all the classes in each grade were randomly selected. All the students in the selected classes were included in the survey. The questionnaire contained a rich array of information for these students, including basic information, family background, study conditions, psychosocial situations, and exposure to damage during the earthquake. To exploit school transitions to facilitate identification, only the 7th, 8th, 10th, and 11th grades in Wenchuan County were included in the analysis. Additionally, although the survey required students to self-report their test scores, the research team collected administrative test scores from 2008 to 2010. Administrative scores were used to evaluate students' academic performance.

In addition to student questionnaires, the research team collected data on schools. Given that the survey occurred two and a half years post-earthquake, during which schools in Wenchuan County underwent reconstruction or consolidation, the research team devised and distributed questionnaires to both "old schools" (those in Wenchuan County during the earthquake) and "new schools" (those in Wenchuan County in December 2009 when the survey was conducted).[6] The headmaster or staff member of a new school was tasked with providing

[5] This survey was part of the *Kin Mirai Kadai Kaiketsu Jigyo* project, and the field research was conducted by the Renmin University of China and Sichuan University (Park et al. (2015)).

[6] Henceforth, the terms "old school" and "new school" refer to the schools existing in Wenchuan County at

school information, while a staff member familiar with the overall affairs of the old schools was responsible for completing the information regarding the old schools. Consequently, this survey provides comprehensive information on each student's background characteristics and post-earthquake conditions, along with detailed school and class information spanning the period from the earthquake to three years later.

In Wenchuan County, the epicenter and most severely affected county, approximately 94% of school buildings were damaged. At the same time, the severity of damage varied across schools within the county, providing the variation needed to study how students respond to differing levels of earthquake exposure among their peers. All middle and high schools in Wenchuan County were surveyed and included in the analysis, yielding four schools serving the relevant grades.

3.2 Variables

Table 1 summarizes the statistics of test scores, old school information, and personal background.[7] Panel A reports the full questionnaire sample used in this study, covering 1,278 students in grades 7, 8, 10, and 11 after excluding grades 10 and 11 at the six-year secondary school for which administrative records are unavailable. The completion column shows that 2010 administrative scores are not observed for every student in this full survey sample. Panel B therefore reports the estimation sample of 1,086 students with non-missing 2010 administrative scores. The estimation sample comprises 27 classrooms from 10 school-by-grade groups, with an average class size of 40 students. Missing 2010 administrative outcomes are addressed in a robustness check that imputes missing outcomes using students' self-reported 2010 scores.

3.2.1 Measurement of academic performance

Students' academic performances were assessed using their Chinese and mathematics scores in the 2010 end-of-academic-year examination. This examination, conducted by the county educational bureau, allows for an evaluation of students' relative performance levels within the county. To facilitate the comparability of scores, raw test scores were normalized to have a mean of 0 and a standard deviation of 1. A higher score indicated better performance. Normalized Chinese and mathematics scores were used as academic outcome measures in the subsequent analyses.

different points in time.

[7] The four schools comprise two middle schools and two six-year secondary schools.

Table 1: Summary statistics.

	Completion	Mean	Std. Dev.	Median
Panel A: Full Sample ($N = 1278$)				
Chinese score (2010)	0.88	64.14	17.09	65.00
Math score (2010)	0.85	39.64	25.45	33.00
Standardized Chinese score (2010)	0.88	0.00	1.00	0.10
Standardized math score (2010)	0.85	0.00	1.00	-0.19
Male	1.00	0.50	0.50	1.00
Age	0.99	13.72	1.31	14.00
Ethnic minority	1.00	0.77	0.42	1.00
Tibetan minority	1.00	0.27	0.45	0.00
Parental education	0.99	0.33	0.47	0.00
Earthquake exposure	0.97	0.91	0.29	1.00
Old school damage level	0.97	1.89	0.62	2.00
Panel B: Estimation Sample (Students with Non-Missing Administrative Scores) ($N = 1086$)				
Chinese score (2010)	1.00	63.60	16.87	65.00
Math score (2010)	1.00	39.56	25.40	33.00
Standardized Chinese score (2010)	1.00	-0.03	0.99	0.10
Standardized math score (2010)	1.00	-0.00	1.00	-0.19
Male	1.00	0.48	0.50	0.00
Age	0.99	13.69	1.30	14.00
Ethnic minority	1.00	0.77	0.42	1.00
Tibetan minority	1.00	0.28	0.45	0.00
Parental education	0.99	0.31	0.46	0.00
Earthquake exposure	0.97	0.91	0.29	1.00
Old school damage level	0.97	1.90	0.63	2.00

Notes: Administrative test score records for grades 10 and 11 at one six-year secondary school are unavailable; these students are excluded from both panels. Test scores are standardized to mean zero and unit standard deviation. Ethnic minority equals 1 if the student is non-Han, and 0 otherwise. Tibetan minority equals 1 if the student is of Tibetan ethnicity; this indicator is used as the ethnicity control in all regression specifications. Parental education equals 1 if at least one parent completed high school or above. Old school damage is a five-point scale: 1 (Intact), 2 (Moderate Damage), 3 (Severe Damage), 4 (Partial Collapse), 5 (Full Collapse). Earthquake exposure = $\mathbb{1}(\text{old school damage} \geq 4)$. Psychosocial outcome summary statistics and scale-construction details are reported in Appendix Table A2.

3.2.2 Measurement of psychosocial outcomes

The survey collected information on four psychosocial scales. The Center for Epidemiologic Studies Depression Scale (Radloff, 1977) (CES-D) measures depressive symptoms; higher scores indicate more severe depression. The Strengths and Difficulties Questionnaire (Goodman, 1997) (SDQ) captures behavioral and peer difficulties; higher scores indicate greater difficulties. The Rosenberg Self-Esteem Scale (Rosenberg, 1965) measures self-esteem; in this study, the scale is reverse-coded so that higher scores indicate lower self-esteem. The Family Environment Scale (Moos and Moos, 1981) (FES) measures the quality of the family environment; higher scores indicate more family conflict.

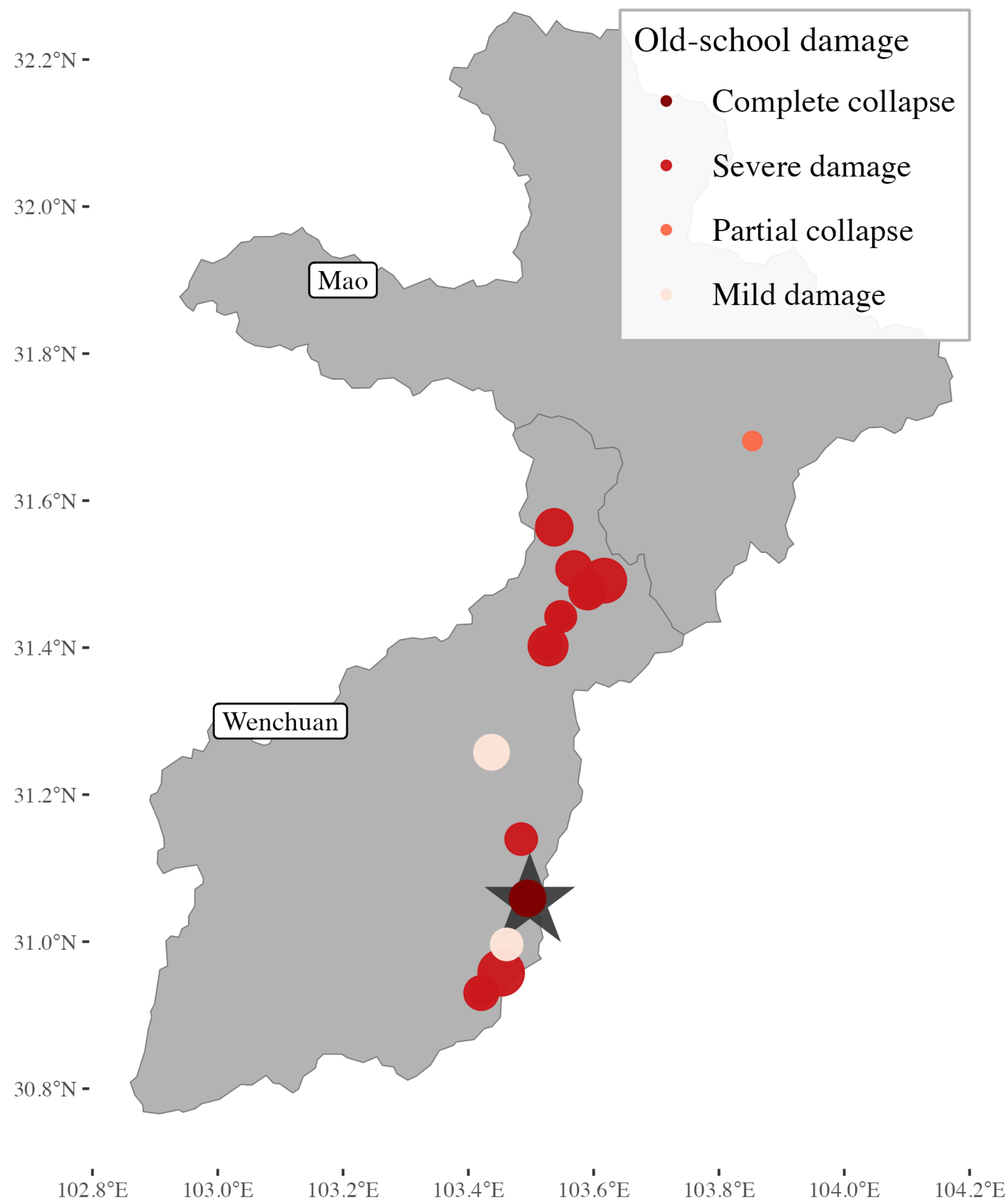
The survey instrument includes a subset of items from each scale. For the CES-D, 12 of the original 20 items are available. For the SDQ, the survey covers only the conduct and peer problems subscales (10 items), excluding the emotional symptoms, hyperactivity, and prosocial subscales. For the Rosenberg scale, all 10 items are included. For the FES, 18 items are available. Each composite score is constructed by summing the available items and normalizing to mean zero and unit standard deviation. Summary statistics are reported in Appendix Table A2.

3.2.3 Measurement of earthquake exposure

I employed the old school's damage level to evaluate the earthquake intensity experienced by each student, which was the independent variable of interest. Most students were on lunch break inside the buildings of their former schools when the earthquake hit Wenchuan County. Consequently, students directly experienced the collapse of school buildings during the earthquake.

The headmasters of the earthquake-hit schools were requested to assess the extent of damage incurred on each floor within their school buildings during the disaster. The condition of the first floor was chosen as the indicator of the overall damage to the school, as damage typically initiates at the lowermost levels of a building and progresses upward. Damage levels on the first floor were classified according to a 5-point scale following official guidelines: "Intact," "Moderate Damage," "Severe Damage," "Partial Collapse," and "Full Collapse."

Figure 2: Map of the old school location.



Notes: A large gray star marks the earthquake epicenter and is drawn behind the nearby school marker. Circles mark old schools from which surveyed students came: darker circles indicate more severe old-school damage, and larger circles indicate more students from that school.

In Figure 2, each dot represents an old school in the two surveyed counties before the earthquake. The size of the dot denotes the number of students enrolled in that school at the time of the earthquake. Most surveyed students were enrolled in old schools in Wenchuan County. The remaining students came from Mao County or schools in neighboring counties or cities within Sichuan Province. The darkness of each dot in Figure 2 corresponds to the severity of the old school damage, with darker shades indicating more extensive damage.

One notable concern regarding the evaluation of outcomes is the potential for reporting errors in the classification of old school damage by the school headmasters. To address this concern, the *Local Earthquake Relief Reports on Sichuan Education System* were cross-referenced.[8] As shown in Figure 2, only two schools were documented as having experienced a full collapse. These two schools were located in the epicenter of Yingxiu Town, which was subjected to the most intense seismic activity. The remaining schools in Wenchuan County, except for the two primary schools, were reported to have suffered severe damage or partial collapse. The apparent anomaly of these two primary schools is that, despite their proximity to the epicenter, contemporaneous news reports indicate that their buildings remained relatively unscathed because they had been built more recently and to a higher structural standard. Appendix Section A presents these reports in more detail. This aligns with official records indicating that Wenchuan County was an extremely severely affected area, with approximately 94% of the school buildings damaged.

Another concern is that students did not necessarily come from old schools in the surveyed counties, in which case, school damage was not recorded.[9] The *Report on the Assessment of the Scope of the Wenchuan Earthquake Disaster* (Government of China, 2008) was used to evaluate the most probable earthquake intensity encountered by these students. Specifically, this report categorizes counties into different earthquake-affected levels: extremely severely affected, severely affected, and affected counties. Matching information on the geographic location of the old schools with the earthquake-affected level of each county showed the extent to which these old and unsurveyed schools were most likely damaged. Through this methodology, 247 additional missing values for old school damage were encoded, with only 70 observations left missing because of unclear school names or lack of earthquake intensity information in that region.

To further mitigate concerns about potential measurement issues, the earthquake exposure index was transformed from a 5-point outcome scale to a binary variable indicating whether the old school building had collapsed (i.e., reported as either partially or fully collapsed). School principals may be unable to accurately recall the precise degree of damage to their old school buildings. However, recording whether an old school building collapses and becomes unusable is a more straightforward task. Therefore, I used the school collapse dummy, rather

[8]It is titled *Sichuansheng Jiaoyu Xitong Wenchuan Teda Dizhen Kangzhen Jiuzai Zhi* in Chinese. This report contains records of the extent of damage to school buildings in each county, as maintained by the Sichuan Education Bureau. This report is part of the *Earthquake Relief Reports* collection, which Cao (2022) uses to assess building damage following the Sichuan earthquake.

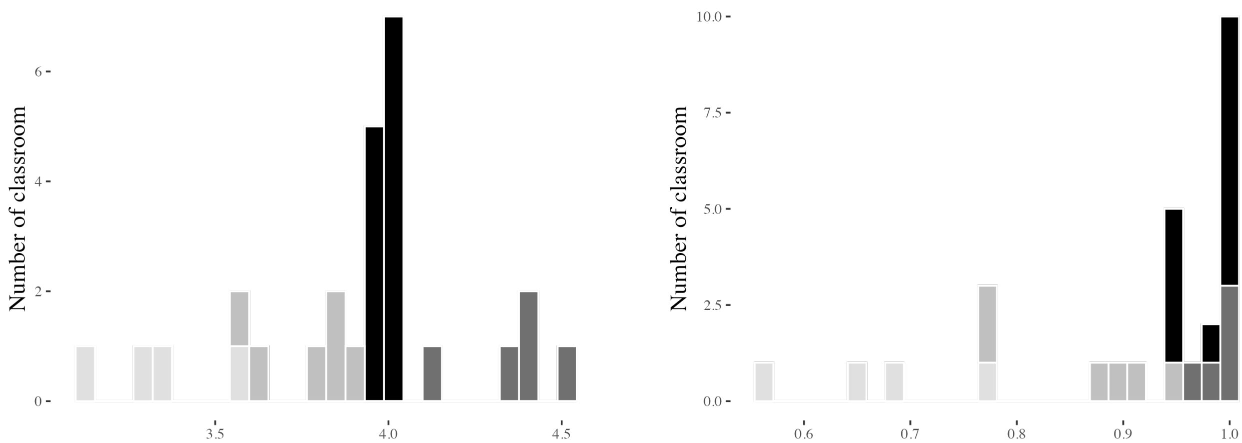
[9]Out of 1,841 surveyed students, 17% reported an old school name that was not registered among the surveyed schools.

than the raw 5-point scale, to indicate earthquake exposure.

3.2.4 Distribution of earthquake exposure

Figure 3 presents the distribution of the two treatment measures across classrooms in each school, where Figure 3a shows the classroom average of the original five-point old-school damage scale and Figure 3b shows the classroom average of the corresponding binarized earthquake-exposure measure. Different colors represent different schools. Among the four schools, two predominantly enrolled students from Wenchuan County, whereas the other two attracted a more diverse student population from adjacent counties. Consequently, the former two schools exhibited limited variations in earthquake exposure levels across classrooms, in contrast to the broader variations observed in the latter two schools.

Figure 3: Distribution of treatment measures.



(a) Classroom average old school damage. **(b)** Classroom average earthquake exposure.

Notes: Earthquake exposure = $\mathbb{1}$(Old school damage level $\geq$ 4). Different colors represent different schools.

4 Identification Strategy

This section presents the empirical strategies and corresponding identification assumptions. Section 4.1 introduces the specification used to identify the effect of peer exposure, and Section 4.2 explains classroom assignment rules and empirically tests whether the assignment induced sorting on predetermined characteristics, which could bias the peer-effect estimates.

4.1 Model specification

To estimate the relationship between classroom peer exposure and test scores, I use a fixed-effects model that exploits differences in the share of school-collapsed classmates across classrooms within the same school, grade, and academic track. This design compares students who entered the same school-grade group but were assigned to classrooms with different levels of peer exposure. The estimating equation is

$$y_{sglri} = \beta_1 \cdot \overline{exposure}_{sglri} + \beta_2 \cdot exposure_{sglri} + \boldsymbol{C}_{sglri}\boldsymbol{\gamma} + \eta_{sg} + \epsilon_{sglri}, \tag{1}$$

where y_{sglri} denotes the 2010 Chinese score, math score, or total score of student i in classroom l, grade g, school s, and academic track r. The term $\overline{exposure}_{sglri}$ is the leave-one-out share of classmates whose old school collapsed,[10] while $exposure_{sglri}$ indicates whether student i's own old school collapsed. The control vector $\boldsymbol{C}_{sglri}$ has coefficient vector $\boldsymbol{\gamma}$ and includes demographic characteristics, class size, and an academic track indicator; when the outcome is a test score, pre-enrollment test scores and a missing-score indicator are also included. Demographic characteristics are gender, age, and Tibetan minority. The fixed effect η_{sg} is a school-grade FE. The error term is ϵ_{sglri}. The coefficient β_1 measures the relationship between classroom peer exposure and the student's test score, whereas β_2 measures the relationship between own school collapse and the student's test score.[11]

The identifying variation comes from within-school-grade differences in peer exposure, or within-school-grade-track differences where track is implemented, after netting out own exposure. The specification therefore relies on comparisons across classrooms within the same school-grade cell, or school-grade-track cell, rather than on comparisons across schools or cohorts. Including own exposure ensures that β_1 is identified net of a student's own disaster experience. I report the peer effects with fixed effects only, and also with additional controls. Standard errors are clustered at the classroom level. Because the sample contains only 27 classrooms, wild-cluster bootstrap inference is implemented as well[12] (Cameron and Miller, 2015).

[10] Throughout the paper, peer exposure is measured as a leave-one-out classroom average, following the standard peer-effects framework in Manski (1993).

[11] In the terminology of Manski (1993), this analysis is reduced-form in the sense that it does not separately identify whether classmates' earthquake exposure affects student i directly through classmates' predetermined characteristics, an *exogenous effect*, or indirectly through classmates' contemporaneous outcomes, an *endogenous effect*.

[12] Following Cameron and Miller (2015), I implement a two-point Rademacher wild cluster bootstrap at the classroom level. For each coefficient under test, I first estimate the restricted model that imposes the null hypothesis, keep the regressors fixed, assign each classroom a common weight of -1 or 1, construct bootstrap outcomes by multiplying the restricted-model residuals by those classroom-level weights, and then recompute the clustered t-statistic. Reported bootstrap p-values are based on 999 replications.

4.2 Addressing concerns of unobservable variables

For the coefficient on peer exposure to admit a causal interpretation, classroom peer composition must not be correlated with unobserved student characteristics in ways that would confound identification. The main concern is that a performance-based assignment rule could generate classroom-level ability clustering, in which case the estimated peer effect would partly reflect sorting on academic ability rather than spillovers from classmates' earthquake exposure.

School interviews indicate that classroom assignment followed a rank-based allocation rule that was tied to entrance-exam rankings but still dispersed students across classrooms rather than sorting them into high- and low-performing classes. In the regular track, students were not assigned by placing all top-ranked students in one class and all lower-ranked students in another. Instead, schools typically rotated students across classrooms in rank order: for example, rank 1 was placed in Class 1, rank 2 in Class 2, rank 3 in Class 3, and rank 4 returned to Class 1, with the same pattern repeated down the ranking. This rule spreads students of different achievement levels across regular classrooms and therefore does not mechanically generate ability clustering across those classes. In some schools, the highest-ranked students were first separated into an advanced track; the remaining students were then allocated across regular classrooms using the same rank-based rule. The identifying assumption is therefore that, within school, grade, and class track, classroom allocation is plausibly as good as random with respect to unobserved determinants of later outcomes.

To assess this implication empirically, I conduct pairwise balance tests in the spirit of Guo and Qu (2022). Within each school-grade-track cell, I form all feasible classroom pairs, which yields 21 classroom pairs in total. Because the student's own old-school collapse status is the underlying treatment variable in this setting, I first examine whether that variable is balanced across classrooms. Neither the collapse indicator nor the raw five-point old-school damage scale yields any rejection at the 5% significance level in the pairwise t-tests or pairwise KS tests. This is a direct evidence that classrooms were not formed on earthquake exposure itself.

I then turn to other demographic characteristics: age, gender, Tibetan minority, parental education.[13] Among these and the old-school damage scale, 7.0% of the pairwise t-tests reject at the 5% significance level. This rate is close to the false-positive benchmark implied by repeated testing.[14] Combined with the treatment-variable results above, this evidence is consistent with an as-good-as-random allocation rule within school-grade-track cells rather than deliberate classroom formation on earthquake exposure. Appendix Table A3 and Figure A2 report the full pairwise balance results.

[13] The two main non-Han groups in the survey area are Qiang and Tibetan. Tibetan minority is used as the ethnicity control in all regression specifications. For the rejection-rate calculation, I include the old-school damage scale but exclude the binary collapse indicator because it is mechanically derived from that scale.

[14] Across these five characteristics, Table A3 reports 100 valid pairwise t-tests. Under exact balance, a 5% test level implies about 5.00 false rejections on average, or 5.0% of tests.

5 Results

This section presents my estimates of the effects of earthquake exposure, both own exposure and the classroom-average peer exposure, on students' developmental outcomes. I first estimate the effects on academic outcomes, quantified through standardized test scores. Next, I estimate the effects on psychosocial and behavioral outcomes.

5.1 Academic outcomes

Table 2: Effects of own and peer earthquake exposure on test scores.

	Chinese		Math		Total	
	(1)	(2)	(1)	(2)	(1)	(2)
Own exposure	−0.30***	−0.33***	−0.32*	−0.38**	−0.37**	−0.41**
	(0.01)	(0.00)	(0.06)	(0.02)	(0.03)	(0.01)
Peer exposure	−3.94**	−4.43**	−2.23	−3.31**	−3.12**	−4.01***
	(0.03)	(0.02)	(0.12)	(0.02)	(0.04)	(0.01)
School-grade FE	Yes	Yes	Yes	Yes	Yes	Yes
Track FE	Yes	Yes	Yes	Yes	Yes	Yes
2008 test score controls	Yes	Yes	Yes	Yes	Yes	Yes
Demographic & class controls	No	Yes	No	Yes	No	Yes
Adjusted R^2	0.20	0.29	0.22	0.24	0.33	0.36
Number of observations	1046	1046	1046	1046	1046	1046

Notes: The sample includes students from both collapsed and non-collapsed old schools, subject to non-missing outcomes. Peer exposure is the leave-one-out classroom share of students whose old school collapsed; own exposure is an indicator for the student's own old-school collapse. All scores are standardized with mean 0 and standard deviation 1. Column (1) includes school-grade FE and a track indicator only. Column (2) adds the relevant 2008 test score, a missing-score indicator, age, gender, Tibetan minority, and class size. Wild cluster bootstrap p-values, based on 999 replications, are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 2 shows that the main result is strongest for Chinese. In column (2), the coefficient on peer exposure is −4.43 for Chinese (p = 0.02). Since peer exposure is measured as a leave-one-out classroom share, this coefficient indicates that a 1 percentage point increase in the share of school-collapsed classmates lowers Chinese scores by about 0.04 standard deviations. In a 40-student class, one additional severely affected student raises the leave-one-out peer-exposure measure by about 1/39, which implies a decline of roughly 0.11 standard deviations in Chinese score. The coefficient remains stable with or without the additional controls, which suggests that the result is not driven by the specific control set. The peer-exposure coefficients for Total score are likewise negative and statistically significant under bootstrap inference (p = 0.01), while Math is also negative and statistically significant (p = 0.02).

I benchmark our estimated magnitude against the displacement literature. In a Hurricane Maria setting, Özek (2023) reports that a 1 percentage point increase in the share of hurricane migrants lowers averaged reading and math scores by about 0.8% of a standard deviation. In a violence-displacement setting, Padilla-Romo and Peluffo (2023) reports a composite-score decline of about 0.4% of a standard deviation per 1 percentage point increase in the share of violence-displaced peers, with reading- and math-specific effects statistically indistinguishable. These comparisons should be interpreted with some caution because they are not exact apples-to-apples benchmarks. This study measures peer exposure at the classroom level, whereas those papers typically define peer shares at the school-by-grade level. A classroom-based measure likely captures more intensive day-to-day interaction, so some difference in magnitude is expected even before turning to differences in setting. Even with that caveat, however, the Chinese peer-exposure coefficient in column (2) remains large: it implies a decline of about 4% of a standard deviation for each 1 percentage point increase in the share of school-collapsed classmates. One plausible reason is that those papers study spillovers from affected newcomers onto unaffected incumbents, whereas the present paper examines an intragroup setting in which all students were affected by the earthquake and 86% experienced school collapse. The coefficient on own exposure is also significantly negative for all subjects, indicating that students whose own school collapsed likewise score lower, independent of their peers' exposure.

5.1.1 Robustness checks

Appendix Figure A3 reports five robustness checks for Table 2. The first check replaces missing 2008 test scores with the subject-specific median while retaining the missing-score indicator. The second replaces the collapse dummy with a three-level exposure measure based on old-school damage severity. The third restricts the sample to students whose old school is observed in the survey data, thereby excluding the cases for which earthquake exposure is inferred from external records. The fourth restricts the sample to schools with greater within-school variation in exposure. The fifth addresses potential sample-selection concerns from missing 2010 administrative outcomes by replacing missing raw 2010 Chinese and math scores with the corresponding self-reported raw scores when available, then re-standardizing the combined outcomes and re-estimating the baseline specification.

The overall pattern again most clearly supports a negative peer effect for Chinese. The peer-exposure coefficient remains negative in all five checks. In the imputed-outcome exercise, the sample expands from 1,046 to 1,508 students, and the peer-exposure coefficients remain negative for Chinese, Math, and Total. These imputed-outcome estimates are smaller in magnitude than in the administrative-only sample, which is plausible given the moderate correlation between self-reported and administrative scores and the resulting measurement noise. Even so, the broader-sample exercise indicates that the adverse peer effect is not an artifact of restricting the analysis to students with non-missing administrative outcomes. Overall, the appendix robustness checks continue to show that the evidence for a negative peer effect is strongest for Chinese scores.

I also complement these exercises with the coefficient-stability analysis based on Oster (2019), reported in Appendix Table A1. That test asks whether unobservables would need to be selected to an implausible degree relative to observables to explain away the estimated peer effect. The appendix results continue to show that the peer-exposure coefficient is stable in the controlled specification.

5.1.2 Placebo test

I next conduct a placebo classroom-assignment exercise to assess whether the estimated peer effect could arise from incidental classroom composition rather than from the actual share of school-collapsed classmates. Within each school-grade cell, I randomly reassign students across the observed classrooms, preserving the number of classrooms and their class sizes. For each reassignment, I recompute peer exposure and re-estimate the baseline fixed-effects specification for Chinese and total scores. This procedure therefore asks whether a coefficient of the observed magnitude would also appear under arbitrary within-school-grade classroom composition.

Figure 4 shows that it does not. Across 50 placebo reallocations, the peer-exposure coefficients are centered near zero for both outcomes, whereas the observed estimates lie far in the negative tail of the placebo distribution. None of the placebo reallocations generates a coefficient as large in absolute value as the observed estimate for either Chinese or total score. This pattern is consistent with the interpretation that the main effect reflects the actual classroom share of school-collapsed peers, rather than accidental grouping within school-grade cells.

Appendix Table A4 complements this exercise by treating predetermined student characteristics as outcomes in the whole-sample regression specification. None of the peer-exposure coefficients in that placebo table is statistically significant at the 5 percent level.

5.2 Psychosocial Conditions and Subjective Wellbeing

Here I examine the effect of peer exposure on students' psychosocial outcomes and subjective wellbeing. This section asks whether peer exposure is associated with psychosocial conditions, peer relationships, and subjective wellbeing within a setting where classmates shared the disaster experience. To assess this possibility, this section reports estimates for broad psychosocial scales, SDQ subscales, and wellbeing outcomes measured through student self-reports in the survey. The outcomes include the CESD depression index, the SDQ difficulties index, the

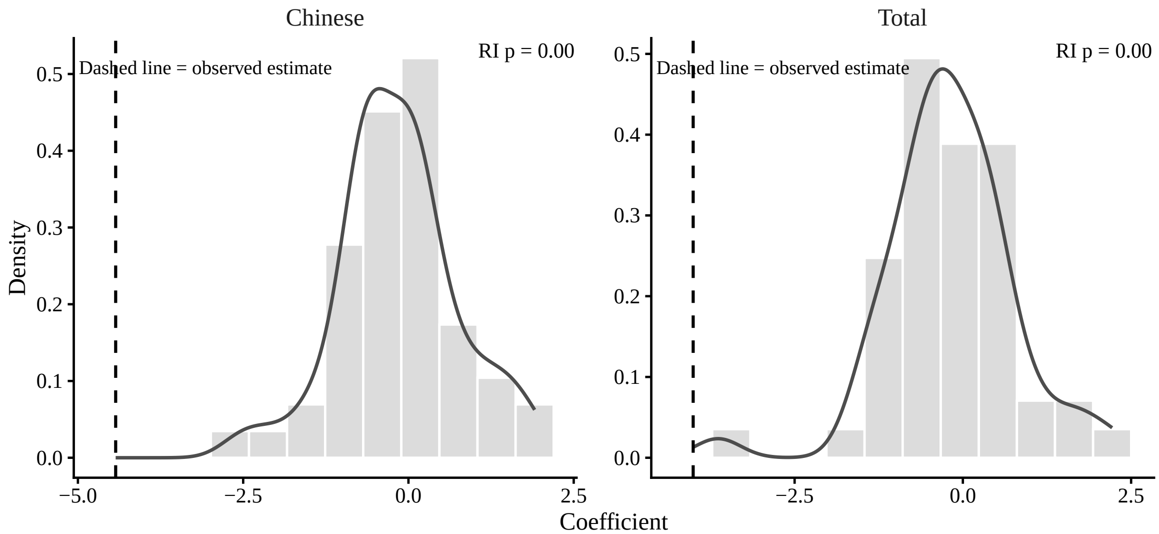


Figure 4: Plausible classroom-assignment test for peer exposure.

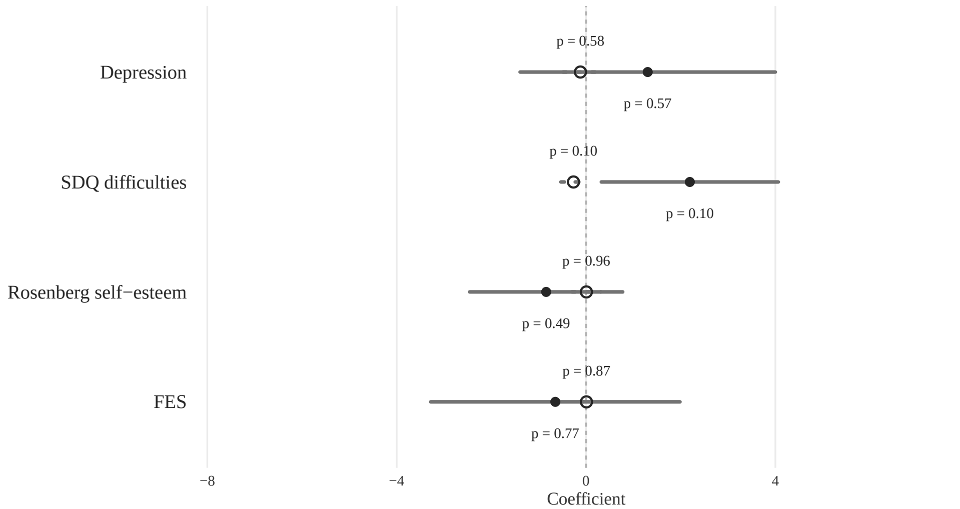
Notes. Each panel reports the placebo distribution of the peer-exposure coefficient from 50 random reallocations of students across observed classrooms within the same school-grade cell. The regression specification matches column (2) of Table 2. The dashed vertical line marks the observed estimate from the actual data. The randomization-inference p-value is the share of placebo estimates whose absolute value is at least as large as the observed estimate.

Rosenberg self-esteem scale, the FES family environment scale, subjective happiness, self-rated health, and study motivation.

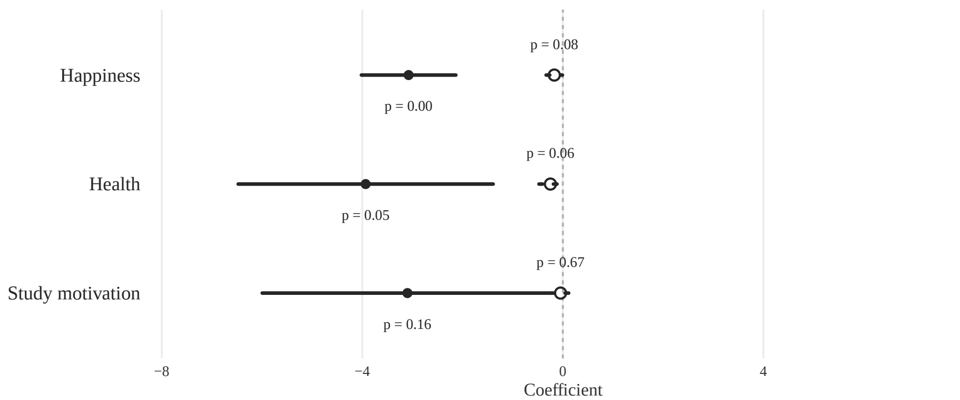
Figure 5 presents the main results for these outcomes. The own exposure coefficient is estimated with empirical model 1. The peer-effect estimates are obtained by restricting the sample to school-collapsed students and estimating:

$$m_{sglir} = \theta_1 \cdot \overline{exposure}_{sglir} + \boldsymbol{X}_{sglir}\boldsymbol{\rho} + \eta_{sg} + \nu_{sglir}, \tag{2}$$

where $\boldsymbol{X}_{sglir}$ includes age, gender, Tibetan minority, and class size. The coefficient θ_1 captures whether, among students whose own school collapsed, differences in classroom peer exposure are associated with differences in non-academic outcomes.



(a) Psychosocial outcomes



(b) Wellbeing and motivation

Figure 5: Peer exposure, psychosocial outcomes, and wellbeing.

Notes. Each panel reports effect of peer's earthquake exposure and own exposure. Filled markers (solid line) report the effect of peer exposure, measured as the share of classmates whose old school collapsed. Hollow markers (dashed line) report the effect of own exposure. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. P-values are based on wild cluster bootstrap inference. Controls: school-grade FE, track indicator, age, gender, Tibetan minority, and class size. In panel a, higher values indicate worse psychosocial conditions. In panel b, higher values indicate better outcomes.

5.2.1 Psychosocial outcomes

Figure 5 begins with four standardized psychosocial composites: CESD depression, SDQ difficulties, Rosenberg self-esteem, and family environment. Higher values indicate worse outcomes for all four measures.

Panel a shows little evidence that more severe earthquake exposure further worsens depression, self-esteem, or family environment within the disaster-affected sample. Among the four psychosocial composites, the peer-effect point estimate is again largest for the SDQ summary measure, which worsens by 2.19 standard deviations as the share of school-collapsed classmates increases and is borderline significant under bootstrap inference ($p = 0.10$). By contrast, the own-exposure estimate for the SDQ composite points toward an improvement rather than a deterioration, but it is likewise only borderline significant ($p = 0.10$). Overall, the psychosocial-composite evidence does not isolate a broad deterioration in depression, self-esteem, or family environment; if anything, it points more narrowly toward classroom difficulties captured by the SDQ block.

This null pattern should be interpreted carefully, however. The comparison here is within a disaster-affected sample, and even the baseline group consists of students who were heavily exposed to the earthquake but did not experience school collapse. The results therefore indicate only that more severe exposure, or a higher share of more severely exposed classmates, does not further deepen depression, self-esteem loss, or family-environment deterioration within this already affected population; they should not be read to mean that depression or other psychosocial distress was absent in the disaster-affected sample as a whole.

5.2.2 Wellbeing and motivation

Panel b reports subjective wellbeing and study motivation. The survey records self-reported happiness and self-reported health on recoded raw scales, with higher values indicating better outcomes. Subjective wellbeing declines under both peer exposure and own school collapse. Happiness is lower in classrooms with more school-collapsed classmates ($p < 0.01$), and it is also lower for students whose own school collapsed, though the direct-effect estimate is only marginally precise under bootstrap inference ($p = 0.08$). Self-reported health shows the same pattern: it declines under peer exposure ($p = 0.05$) and is also lower for students whose own school collapsed ($p = 0.06$). The figure also reports a self-reported study-motivation item, recoded so that higher values indicate stronger motivation. Study motivation declines under peer exposure, but the estimate is not statistically sharp under bootstrap inference ($p = 0.16$), while the own-exposure estimate is close to zero ($p = 0.67$). Overall, the pattern in Panel b is one of worse subjective wellbeing under both peer exposure and own school collapse, with suggestive but less precise evidence that peer exposure also weakens study motivation.

6 Heterogeneous Effects

This section examines heterogeneity in the estimated effects of earthquake exposure along two dimensions: prior academic rank and gender. Prior literature documents that peer effects are not uniform across the achievement distribution: higher-ranked students are more likely to sustain their academic advantage over time (Murphy and Weinhardt, 2020), are more engaged in constructive competitive and social activities (Cicala, Fryer, and Spenkuch, 2018; Dobrescu et al., 2021; Tincani, 2025), and receive more adult attention and effort (Carneiro et al., 2025). For gender, prior work likewise suggests that peer effects may differ between female and male students (Lavy and Schlosser, 2011).

Throughout, I use the same interaction design for both dimensions of heterogeneity. Let Z_i denote the relevant group indicator, where $Z_i = H_i$ for the prior-rank analysis and $Z_i = G_i$ for the gender analysis. For own exposure, I estimate

$$Y_{isgc} = \alpha + \beta D_i + \gamma Z_i + \delta\left(D_i \times Z_i\right) + \rho P_i + X_i'\theta + \eta_{sg} + \varepsilon_{isgc}, \tag{3}$$

and for peer exposure, I estimate

$$Y_{isgc} = \alpha + \beta P_i + \gamma Z_i + \delta\left(P_i \times Z_i\right) + X_i'\theta + \eta_{sg} + \varepsilon_{isgc}. \tag{4}$$

The reference group is low-rank students in the rank analysis and male students in the gender analysis. In each panel, the first row reports $\hat{\beta}$, the estimated effect for the reference group. The second row reports $\hat{\beta} + \hat{\delta}$, the estimated total effect for the comparison group (high-rank, or female). The third row reports $\hat{\delta}$, the interaction term, which measures the differential effect between the comparison group and the reference group.

6.1 Heterogeneity by prior academic rank

6.1.1 Rank group definition

Baseline rank is constructed from students' test scores in the final year of their old school, observed in 2008. These scores are observed after the earthquake but before students enter their new schools and new classrooms, so baseline rank is predetermined with respect to the subsequent peer composition. The remaining concern is therefore a contamination problem rather than a peer-composition problem: if severe old-school damage already depressed a student's 2008 score, then heterogeneity by "baseline rank" could partly sort students by earthquake severity rather than by prior academic standing. I address this concern in two ways. First, regressions of 2008 achievement on old-school damage with the standard controls and school-grade FE show no evidence that severe damage reduced baseline performance; for example, the own-exposure coefficient is 0.10 standard deviations for the 2008 total score ($p = 0.38$), and it is likewise small and statistically insignificant for 2008 language, math, and baseline rank. Second, I define rank using buffered extreme groups rather than a hard top-versus-bottom split. For the buffered school-grade-rank definition used in the subsequent analysis, I classify the bottom 40 percent as low-rank and the top 40 percent as high-rank, leaving the middle 20 percent unlabeled. This buffer keeps the comparison focused on clearly lower-ranked and clearly higher-ranked students rather than observations close to the rank cutoff.[15] The 40 percent cutoff is an arbitrary choice, and I assess its sensitivity by varying the buffered threshold from 35 to 45 percent; the corresponding estimates are reported in Appendix Figure A4.

6.1.2 Results for academic outcomes

Panel A of Table 3 shows that the own-exposure academic losses are strongest for low-rank students. Chinese scores fall by 0.81 standardized units ($p < 0.01$), math scores fall by 0.70 standardized units ($p = 0.01$), total scores fall by 1.45 standardized units ($p < 0.01$), and overall rank percentile falls by 0.27 ($p < 0.01$). The corresponding estimates for high-rank students are close to zero, so the high-minus-low contrasts remain sizable: 0.70 for Chinese ($p = 0.10$), 0.74 for math ($p = 0.03$), 1.35 for total score ($p = 0.04$), and 0.31 for overall rank percentile ($p < 0.01$).

Panel B of Table 3 shows that peer-exposure academic losses are most evident among low-rank focal students. Chinese scores fall by 4.93 standardized units ($p = 0.01$), math scores fall by 2.57 standardized units ($p = 0.10$), total scores fall by 3.44 standardized units ($p = 0.02$), and overall rank percentile falls by 1.24 points ($p = 0.03$). The corresponding high-rank focal estimates are smaller and statistically insignificant, and the high-minus-low contrasts are sizable: 2.83 for Chinese ($p = 0.02$), 2.30 for math ($p = 0.02$), 2.40 for total score ($p = 0.01$), and 1.28 for overall rank percentile ($p < 0.01$). Appendix Figure A4 shows that these peer-exposure estimates are unaffected by the choice of the 40/40 rank threshold. Lower-rank students are more affected by earthquake exposure, both directly and through peer influence.

6.2 Heterogeneity by gender

6.2.1 Results for academic outcomes

The central finding in this subsection is that the gender difference in earthquake exposure effects is driven entirely by peer exposure, not by own exposure. Male and female students experience indistinguishable losses from their own school collapse, but female students bear a significantly larger negative peer effect than male students.

Panel C of Table 3 shows that own-exposure effects are similar across gender. Male students see Chinese scores fall by 0.40 standardized units ($p < 0.01$), total scores fall by 0.86

[15] As a simple back-of-the-envelope calculation intended to show that false classification is very limited under the buffered group definition, I assume that the effect of own exposure on test scores is 0.41 standard deviations, as estimated in Table 2, add 0.41 back to the 2008 total score of students whose own old school collapsed, and then re-rank students. Under this adjustment, no students under the school-grade 40/40 split move from the calibrated high-rank group into the observed low-rank group.

Table 3: Academic Heterogeneity by Prior Rank and Gender

	Outcome			
	Chinese score	Math score	Total score	Overall rank (pctile)
Panel A: Prior rank heterogeneity, own exposure				
Low-rank	-0.81^{***}	-0.70^{***}	-1.45^{***}	-0.27^{***}
	(< 0.01)	(0.01)	(< 0.01)	(< 0.01)
High-rank	-0.12	0.04	-0.11	0.03
	(0.68)	(0.88)	(0.79)	(0.66)
High minus low	0.70	0.74^{**}	1.35^{**}	0.31^{***}
	(0.10)	(0.03)	(0.04)	(< 0.01)
Observations	548	548	548	548
Panel B: Prior rank heterogeneity, peer exposure				
Low-rank	-4.93^{**}	-2.57	-3.44^{**}	-1.24^{**}
	(0.01)	(0.10)	(0.02)	(0.03)
High-rank	-2.11	-0.27	-1.05	0.04
	(0.23)	(0.88)	(0.42)	(0.91)
High minus low	2.83^{**}	2.30^{**}	2.40^{**}	1.28^{***}
	(0.02)	(0.02)	(0.01)	(< 0.01)
Observations	499	499	499	499
Panel C: Gender heterogeneity, own exposure				
Male	-0.40^{***}	-0.45^{***}	-0.86^{***}	-0.15^{***}
	(< 0.01)	(0.01)	(< 0.01)	(< 0.01)
Female	-0.43^{**}	-0.32	-0.75^{**}	-0.11^{**}
	(0.03)	(0.13)	(0.02)	(0.02)
Female minus male	-0.03	0.13	0.11	0.03
	(0.91)	(0.58)	(0.80)	(0.65)
Observations	702	701	700	700
Panel D: Gender heterogeneity, peer exposure				
Male	-3.48^{**}	-1.67	-3.86^{*}	-0.72^{**}
	(0.04)	(0.35)	(0.07)	(0.03)
Female	-5.09^{**}	-2.58	-6.39^{**}	-1.14^{**}
	(0.01)	(0.14)	(0.01)	(0.02)
Female minus male	-1.61^{***}	-0.90	-2.53^{**}	-0.42^{*}
	(< 0.01)	(0.28)	(0.01)	(0.05)
Observations	634	633	632	632

Notes. This table reports heterogeneity by prior rank and gender. In Panels A and B, the reference group is low-rank students; in Panels C and D, the reference group is male students. Low-rank and high-rank are defined within school-grade cells using the 2008 baseline score distribution: the bottom 40% are classified as low-rank and the top 40% as high-rank, leaving the middle 20% unlabeled. All score outcomes are standardized with mean 0 and standard deviation 1. Overall rank is measured as percentile rank. All specifications include school-grade FE, a track indicator, age, gender, Tibetan minority, and class size. Test-score specifications additionally control for the relevant 2008 test score and a missing-score indicator. Wild cluster bootstrap p-values, based on 999 replications, are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

standardized units ($p < 0.01$), and overall rank percentile fall by 0.15 ($p < 0.01$); female students see comparable declines of -0.43 for Chinese ($p = 0.03$), -0.75 for total score ($p = 0.02$), and -0.11 for overall rank percentile ($p = 0.02$). None of the female-minus-male contrasts is statistically distinguishable from zero.

Panel D shows a clear and significant gender difference in peer-exposure effects. Female students experience noticeably larger losses: -5.09 standardized units for Chinese ($p = 0.01$), -6.39 for total score ($p = 0.01$), and -1.14 for overall rank percentile ($p = 0.02$). The corresponding estimates for male students are smaller: -3.48 for Chinese ($p = 0.04$), -3.86 for total score ($p = 0.07$), and -0.72 for overall rank percentile ($p = 0.03$). The female-minus-male interaction terms confirm the difference is statistically significant: -1.61 for Chinese ($p < 0.01$) and -2.53 for total score ($p = 0.01$). Female students are more negatively affected by peer exposure than male students.

7 Mechanisms

What channels link severely affected students to the decline in their peers' academic performance? Prior literature suggests that several channels may be at work. I first introduce these potential channels and then provide empirical evidence to support the mechanism discussion.

Classroom disruption. The disruption channel runs through peer conduct and misbehavior rather than through direct academic interaction. In Lazear (2001), classroom learning is framed as a joint production process in which disruptive behavior by some students reduces effective learning time for others. Sacerdote (2011) reviews a broad body of evidence showing that peer effects may also arise through behavioral channels such as classroom conduct and the allocation of teacher attention. Along the same lines, Carrell, Hoekstra, and Kuka (2018) and Zhao and Zhao (2021) provide evidence that peers with worse conduct or more disruptive behavior can harm classmates' outcomes.

Peer relationships. Academic performance can be shaped by the quality of peer relationships within the classroom. Students choose their own mode and intensity of social interaction, and the quality of these interactions can shape both the magnitude and the direction of peer effects (Brady, Insler, and Rahman, 2017). The disaster and social-connection literature suggests that individuals exposed to severe shocks, especially those who develop PTSD or PTSS, often experience declines in social participation, prosocial behavior, and social support. Post-disaster PTSD may reduce children's social engagement because affected individuals perceive less support, engage in active avoidance, or become excluded from existing social networks (Beck et al., 2009; Lai et al., 2018). Traumatic life stress may also erode trust (Song, Song, and Malti, 2024). Evidence beyond child settings further shows that post-disaster relocation can weaken social interaction and social cohesion (Hikichi et al., 2017). In Section 5, the strongest non-academic evidence appears in the peer-relationship component of the SDQ and in lower subjective wellbeing, patterns that are consistent with this channel.

Student-teacher relationships. Teachers shape the classroom relational environment through care, trust, familiarity, responsiveness, and individual attention, and these relational inputs are themselves educationally productive: Thijssen, Rege, and Solheim (2022) show that teachers' relationship skills affect student learning, while Hill and Jones (2018) and Hwang, Kisida, and Koedel (2021) provide complementary evidence that repeated student-teacher matches and teacher continuity improve achievement.

Rank competition. Under this channel, students care about their relative standing within the classroom: facing more rivals who can outperform them sharpens the incentive to study harder. What matters is the competitive environment itself, not simply average peer ability. Murphy and Weinhardt (2020) show that a student's ordinal rank in primary school has lasting causal effects on secondary achievement, independent of underlying ability, indicating that students respond to their position in the classroom hierarchy. Tincani (2025) provides direct evidence in a natural disaster setting: changes to the within-classroom distribution of peer disruptions altered competitive incentives and produced achievement effects that varied across the prior performance distribution, pointing to rank competition as a channel through which peers influence each other. The implication is that how many credible competitors a student faces, and whether that number changes, shapes her effort and, ultimately, her academic outcomes.

Teacher pedagogical practice. Student outcomes change when teachers adjust how they teach in response to the classroom they face. Duflo, Dupas, and Kremer (2011) provide canonical evidence consistent with this mechanism, showing that the gains from tracking partly reflect teachers teaching at a level better matched to students. Aucejo et al. (2022) further show that the effectiveness of teaching practices varies with classroom composition, implying that pedagogical practices interact with the peer environment in shaping achievement.

Time allocation and study effort. A student's allocation of after-school study time is an additional channel through which peer composition can shape academic performance. If a classroom contains more severely affected classmates, learning inside the classroom may become less efficient because lessons are more easily disrupted or instruction becomes less effective. Under that mechanism, students may end up spending less effective time engaged with schoolwork, especially in discretionary study outside school, and this decline could contribute directly to academic loss. Zhao and Zhao (2021) document this pathway in Chinese middle schools, where a higher share of disruptive peers is associated with reduced after-school homework hours alongside lower test scores. Barwick et al. (2026) provide complementary evidence at the university level, showing that peer-influenced disruption can shift study habits in ways that have measurable academic consequences. Together, these studies suggest that after-school study time is a meaningful intermediate channel through which peer composition translates into differences in academic performance.

7.1 Classroom Environment

Classroom disruption and interpersonal relationship deterioration both reflect ways in which peer composition can reshape the classroom environment; I analyze them jointly in this subsection. I probe this channel through three complementary pieces of evidence. The first decomposes the SDQ composite into its conduct-problems and peer-problems subscales: the conduct-problems subscale proxies for the disruption channel, while the peer-problems subscale proxies for the interpersonal relationship channel. The second asks whether interaction intensity amplifies the peer-relationship channel. If this channel is operative, peer exposure should have larger effects where students interact more frequently. Students are assumed to interact more frequently with same-gender classmates, so gender serves as a proxy for interaction intensity; peer exposure is decomposed by the gender of the affected classmates and estimated separately for female and male focal students. The third asks whether the groups most affected academically also report weaker relational resources within the classroom. If the peer relationship or student-teacher relationship channels are operative, affected students should report lower received comfort from peers or teachers, or a reduced willingness to disclose problems.

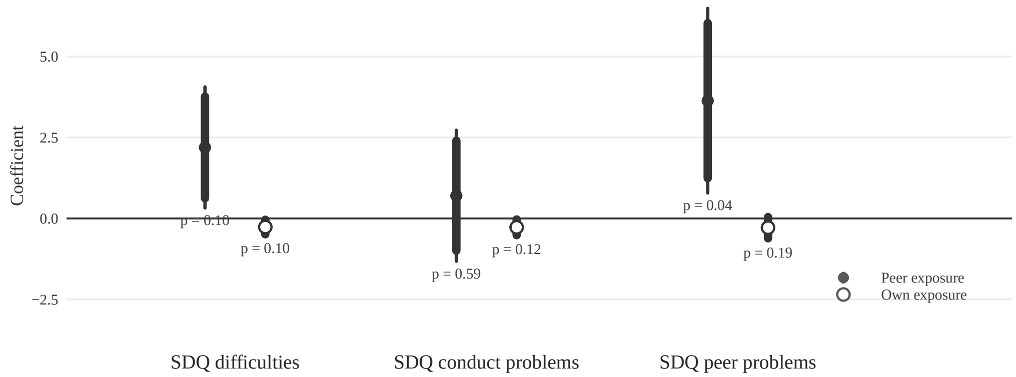
SDQ subscale decomposition. Panel a shows that the conduct-problems subscale responds little to peer exposure, and the item-level evidence in Appendix Figure A5 does not reveal a consistent positive coefficient on any disruption item. This does not support a broad increase in disruptive classroom behavior as the main mechanism. The peer-problems subscale, in contrast, worsens with peer exposure (coefficient $= 3.64$, $p = 0.04$). Item-level increases are most evident in “not liked by peers” and “no good friend,” pointing to deterioration in peer relationships rather than a generalized rise in classroom disorder.

Same- versus other-gender peer exposure. Panel b reports the peer-problems response to same- and other-gender peer exposure, estimated separately for female and male focal students. Female students, who face the larger academic losses shown in Panel D of Table 3, show a pattern consistent with the higher-contact hypothesis: when the share of same-gender school-collapsed classmates rises, peer problems worsen markedly (coefficient = 5.17, $p = 0.04$), while the other-gender estimate is near zero ($p = 0.84$). For male students, the pattern reverses: the peer-problems response loads on other-gender (female) peer exposure (coefficient = 3.32, $p = 0.03$), while same-gender peer exposure is negligible. A natural interpretation is that female students play a more central role in maintaining the relational climate of the classroom; when a larger share of female classmates are severely affected, the relational environment deteriorates in ways that extend to male students as well (Lavy and Schlosser, 2011; Lavy, Paserman, and Schlosser, 2012). Appendix Figure A6 further shows that same-gender peer exposure significantly raises depression among female students, suggesting the channel extends beyond peer relationships to psychological outcomes and may contribute to the larger academic losses reported in Panel D of Table 3. In both cases, the peer-problems increase loads on exposure from female school-collapsed classmates, pointing to an interpersonal channel that operates through relationally salient peer ties rather than a uniform classroom-level shock.

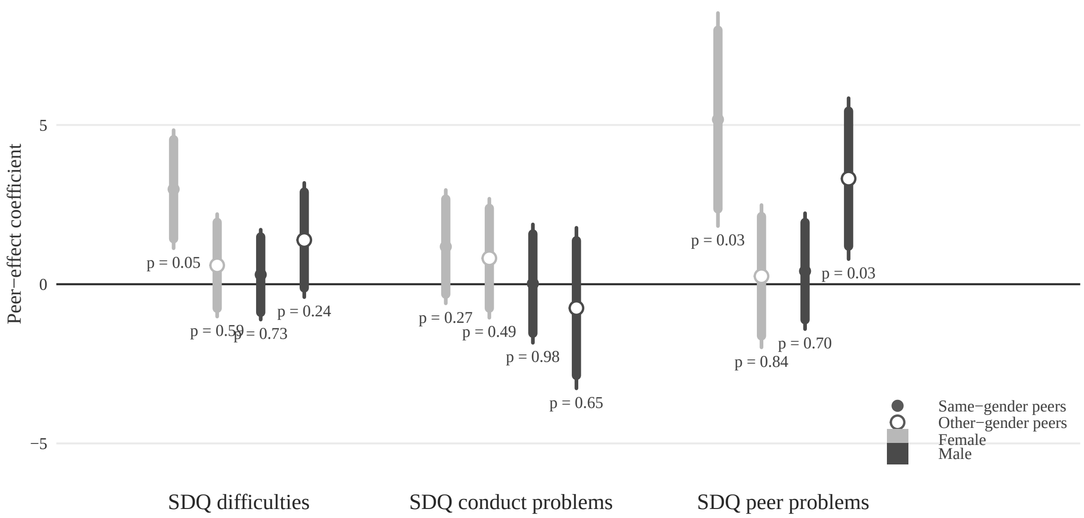
Social support and disclosure. Figure 7 examines outcomes that map directly onto the peer relationship and student-teacher relationship sub-channels. Each channel is measured along two dimensions: (i) received support, captured by whether the student is comforted by peers or by teachers when distressed; and (ii) self-disclosure, captured by whether the student tells peers or teachers about her problems. Panel a reports heterogeneity by prior rank and Panel b reports heterogeneity by gender.

Across rank groups, low-rank students report larger declines in peer comfort (-2.15, $p = 0.02$) and teacher comfort (-1.19, $p = 0.05$) under peer exposure than high-rank students, mirroring the academic rank gradient. For gender, the picture is more nuanced. Under own exposure, female students who themselves experienced school collapse are more likely to disclose to peers (coefficient = 0.18, $p = 0.01$), while received peer or teacher comfort shows no significant change. Under peer exposure, the coefficient on peer comfort for female students is negative (-1.05) and significant under classroom-clustered standard errors ($p = 0.06$) but not under bootstrap inference ($p = 0.20$); teacher comfort and disclosure measures are insignificant.

Across the three pieces of evidence, the classroom environment channel is supported through the **peer relationship** and **student-teacher relationship** sub-channels, and is not supported through the classroom disruption sub-channel.



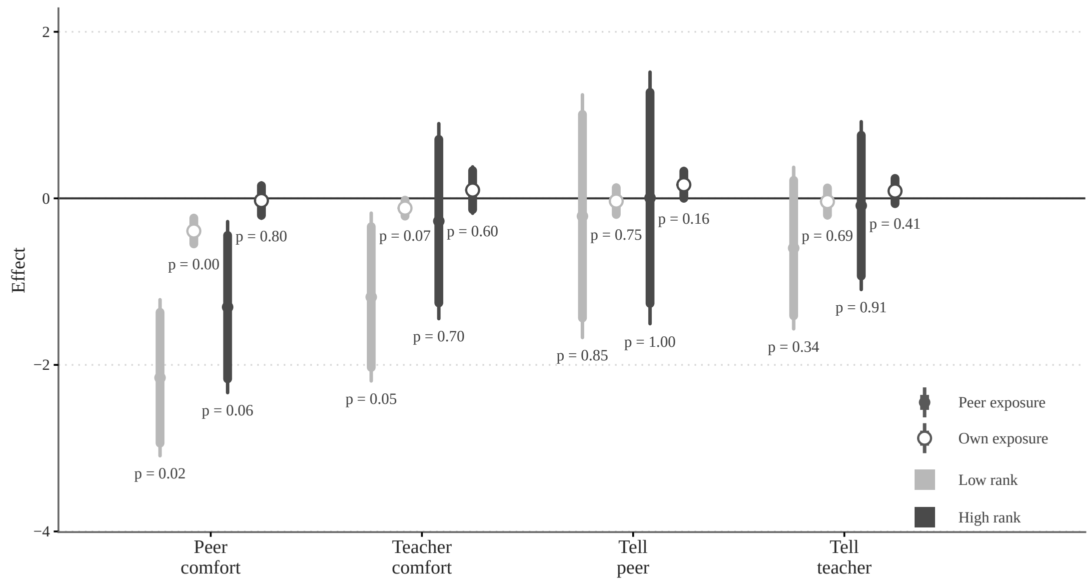
(a) SDQ composite and subscales



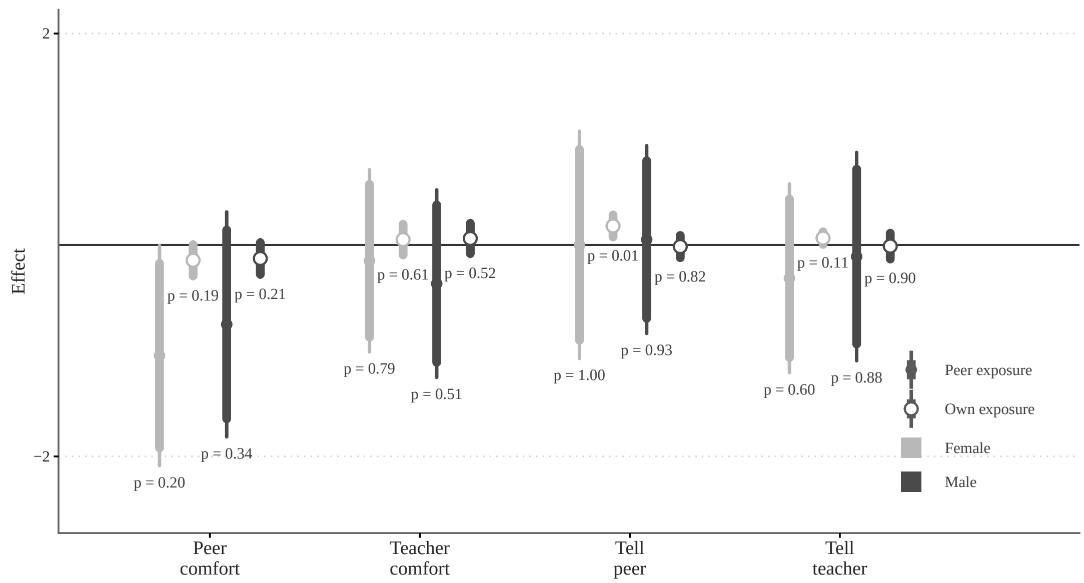
(b) Gender and peer-group decomposition of SDQ peer effects

Figure 6: SDQ decomposition and gender-specific peer-effect patterns.

Notes. Panel a reports the SDQ composite together with the conduct-problems and peer-problems subscales. Filled markers report the effect of peer exposure, and hollow markers report the effect of own exposure. Panel b reports peer-exposure coefficients only, estimated within the sample of students whose own school collapsed. Same-gender peer exposure is the leave-one-out share of school-collapsed classmates of the same gender as the focal student, and other-gender peer exposure is the corresponding share for classmates of the opposite gender. In panel b, lighter elements indicate female students and darker elements indicate male students; filled points denote same-gender peer exposure and hollow points denote other-gender peer exposure. The wider interval reports the 95% confidence interval and the narrower interval reports the 90% confidence interval, based on classroom-clustered standard errors. Labels report wild cluster bootstrap p-values. Controls in both panels are school-grade FE, track indicator, age, gender when applicable, Tibetan minority, and class size. Higher values indicate worse SDQ outcomes.



(a) Rank heterogeneity in support outcomes



(b) Gender heterogeneity in support outcomes

Figure 7: Heterogeneity in support outcomes.

Notes. Panel a reports rank heterogeneity in the four support outcomes from the classroom-interpersonal-relationship mechanism results. Rank is defined by baseline total-score rank within each school-grade cell: high-rank students are the top 40 percent, low-rank students are the bottom 40 percent, and the middle 20 percent are omitted. Panel b reports gender heterogeneity in the same four support outcomes, with female students as the reference group. In both panels, peer effect refers to the coefficient on peer exposure, measured as the share of classmates whose old school collapsed. Own exposure refers to the coefficient on the student's own old-school-collapse indicator; in Panel b, the own-exposure specification controls for peer exposure. Both exposure types are shown with line intervals; filled circles denote peer-effect estimates and hollow circles denote own-exposure estimates. The wider interval reports the 95% confidence interval and the narrower interval reports the 90% confidence interval, based on classroom-clustered standard errors. Labels report p-values from wild cluster bootstrap inference. In Panel a, low-rank estimates are shown in lighter gray and high-rank estimates in darker gray. In Panel b, female estimates are shown in lighter gray and male estimates in darker gray.

7.2 Rank Competition

An alternative is that the peer-exposure results reflect rank competition. When school-collapsed students perform worse, competition for classroom standing becomes less intense. Under this mechanism, students should be able to reduce effort while still preserving, or even improving, their prior rank. This prediction is relevant for both low-rank and high-rank students. If anything, it may be especially salient for high-rank students, because rank brings greater teacher attention, recognition, and rewards, giving them stronger incentives to track changes in competitive pressure.

Beyond that, two other features of the evidence are also inconsistent with this channel. First, overall rank declines rather than improves. The low-rank focal estimate for overall rank is -1.24 ($p = 0.03$), which runs against the prediction that weaker competition should preserve rank. Second, the subject-specific pattern remains inconsistent with rank competition. As shown in Table 1, math has a markedly lower mean (39.64 vs. 64.14) and higher standard deviation (25.45 vs. 17.09) than Chinese, so overall score rank is more sensitive to variation in math performance. Since students compete most directly with similarly ranked classmates (Guo and Qu, 2022),[16] the subject that most determines rank is also the one through which competition is most intense. Rank competition should therefore bite most forcefully through math. If rank competition were driving the peer-exposure effects, math scores should therefore be at least as affected as Chinese scores. In the peer-rank heterogeneity results, Chinese scores for low-rank focal students fall by 4.93 ($p = 0.01$), while the corresponding math estimate is smaller, at 2.57 ($p = 0.10$). Therefore, the evidence does not support the rank competition channel as an explanation for the observed peer effects.

7.3 Teacher Pedagogical Practices

A potential alternative explanation is that teachers endogenously adjust their time allocation in response to classroom earthquake exposure. The matched teacher-survey sample is very small, however, covering only 18 head teachers.[17] I therefore treat the teacher-time evidence as descriptive rather than as stand-alone regression evidence. In those descriptive patterns, head teachers in more exposed classrooms appear to spend more time on student life support, while teaching preparation time shows little corresponding change. The more relevant question is whether these inputs predict achievement. Appendix Table A5 reports positive coefficients on teaching preparation time in both Chinese and math, which is consistent with longer preparation being associated with better academic performance, insignificant though. By

[16]High schools allocate enrollment quotas to middle schools on a hierarchical basis: the best local high schools admit the top-ranked students, the second-best admit the next tier, and so on.

[17]Teachers' self-reported hours for teaching preparation and student support are converted into percentage ranks to mitigate reporting errors and outliers.

contrast, student life-support time shows no evidence of improving achievement. On balance, the teacher-time evidence does not provide support for endogenous adjustment in pedagogical practice as an explanation for the estimated peer effect.

7.4 Time Allocation and Study Effort

A related alternative is that peer exposure lowers academic performance by reducing study effort outside school. If classrooms with more severely affected classmates are less conducive to learning, students may absorb less during class and respond by disengaging from schoolwork more broadly. Under this time-allocation mechanism, Appendix Figure A7 should show lower after-school study time as peer exposure rises. The data do not support this prediction. In the full sample, the peer-exposure estimate for after-school study time is positive and insignificant, and the subgroup results likewise show no evidence of declining study effort. If anything, the strongest pattern appears for high-rank students, whose after-school study time rises with peer exposure (coefficient = 3.51, p = 0.10). This estimate is only marginally significant and should be interpreted cautiously, but its sign runs against the reduced-effort hypothesis.

Overall, the evidence does not support time allocation or reduced study effort as the main mechanism behind the academic loss. If anything, the pattern is more consistent with a compensatory response in which some students, particularly high-rank students, increase study effort outside class when classroom learning becomes less effective. This interpretation is suggestive rather than definitive, but it may help explain why high-rank students do not experience comparable academic declines.

7.5 Teacher Quality, Teacher Assignment, and Classroom Inputs

One concern is that peer exposure may proxy for differences in teacher quality rather than peer effects. Schools that avoided collapse may have had better infrastructure and stronger teachers even before the earthquake, so students arriving from those schools may also have had better prior instruction. Appendix Table A6 addresses this concern by regressing the old-school collapse indicator on the share of high-title teachers in the student's pre-earthquake school. The relationship is close to zero and precisely uninformative ($\hat{\beta} = -0.07$, p = 0.66), which does not support the view that school collapse mainly captures pre-existing differences in old-school teacher quality.

A second concern is selective teacher assignment after school resettlement. If stronger head teachers were disproportionately assigned to classrooms with fewer school-collapsed students, the estimated peer effect could partly reflect teacher sorting rather than peer exposure itself. The matched teacher-survey sample is small, however, covering only 26 classrooms. In that sample, there is no statistically significant difference in head-teacher title between higher- and lower-exposure classrooms: classroom exposure does not predict whether the head teacher holds a high title in either the Chinese sample ($\hat{\beta} = -0.33$, $p = 0.77$) or the math sample ($\hat{\beta} = 0.28$, $p = 0.83$). Given the limited sample, this evidence is necessarily limited, but it does not suggest systematic sorting of stronger teachers into less-exposed classrooms.

The final check asks whether the main estimates are sensitive to directly conditioning on teacher-side inputs. Appendix Figure A8 re-estimates the baseline models after adding, separately and jointly, the share of high-title teachers in the student's old school, the current head teacher's title, and the teacher's teaching-time measure. The peer-exposure coefficient remains negative throughout for Chinese, math, and total scores, and positive throughout for SDQ difficulties. The point estimates move little relative to the baseline specification: the Chinese coefficient ranges from −4.28 to −5.00, the math coefficient from −3.11 to −3.60, the total-score coefficient from −3.79 to −4.46, and the SDQ coefficient from 1.80 to 2.19. Collectively, this evidence suggests that the main peer-exposure results are not driven by old-school teacher quality, selective head-teacher assignment, or these observable teacher-side inputs.

8 Discussion and conclusion

This paper asks whether, after the 2008 Wenchuan earthquake, severely affected students generate spillovers onto other affected classmates once they are reassigned into new classrooms. To the best of my knowledge, it is the first study to estimate intragroup disaster peer effects for the affected students themselves rather than for unaffected incumbent peers. The setting is a post-earthquake school-transition environment in which students from old schools with different damage intensity were recomposed into new classrooms during recovery. Identification comes from within-school-grade differences in the classroom share of school-collapsed classmates under conditionally quasi-random classroom assignment, while controlling for a student's own old-school collapse.

The main results show that peer exposure lowers academic performance, with the strongest and most precise effects on Chinese and total scores and a negative but less precise effect on math. Own exposure is also negatively associated with academic performance. The heterogeneity results show that these losses are not evenly distributed. Peer-exposure effects are largest for low-rank students and for female students, while own-exposure losses are also markedly larger for low-rank students. These patterns indicate that the educational consequences of disaster peer exposure are most evident among students who appear less buffered within the post-disaster classroom environment.

The mechanism evidence points more strongly to deterioration in classroom relationships and interpersonal support than to alternative channels. Peer exposure worsens the peer-problems component of the SDQ, lowers self-reported happiness and health, and is associated with weaker received comfort from peers and teachers, especially for low-rank students. By contrast, the evidence does not support a broad conduct-disruption story: the conduct-problems component of the SDQ is small and insignificant. Nor does it support rank competition, reduced study effort, higher old-school teacher quality, selective teacher assignment, or teacher-side pedagogical adjustment as the main explanation for the academic spillover. The overall pattern is most consistent with a classroom environment in which the relational and wellbeing consequences of severe earthquake experience spill over across affected classmates.

An external-validity question is whether this setting differs from disaster contexts in which school closure itself is a major source of learning loss. In the school-level closure data, the median reported closure is 65.5 days in the raw measure and 60 days in the capped measure. But the earthquake struck in May 2008, just before the summer break, and students had resumed school by September 2008, so part of the measured closure likely overlaps the normal vacation margin rather than active instructional time. This makes the paper closer to rapid-resumption settings such as Andrabi, Daniels, and Das (2021) and Tincani (2025) than to settings such as Morrill and Westall (2023) and Brück, Di Maio, and Miaari (2019), where school closure is more central to the educational shock itself. The estimates here should therefore be interpreted primarily as the effects of earthquake experience and post-disaster peer recomposition, not as reduced-form effects of long school shutdown.

Finally, this paper speaks to one policy-relevant arrangement: what happens when disaster-affected students are placed with one another after school transition. An important alternative policy is to integrate affected students into classrooms dominated by unaffected peers. Existing peer-composition studies are informative about how unaffected incumbent students respond to that arrangement, but they say much less about what happens to the affected students themselves once they are integrated into those classrooms. As a result, this paper cannot provide a welfare comparison between grouping affected students together and integrating them with unaffected peers. A natural next step for future research is therefore to estimate the academic and non-academic consequences of integrated placement for the affected students themselves, so that these alternative post-disaster schooling policies can be compared directly.

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Ethics Statement

The study protocol was reviewed and approved by the Institutional Review Board of the University of Tokyo (No. 15-177).

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Declaration of Competing Interest

I declare that I have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that support the findings of this study are available upon reasonable request from Yasuyuki Sawada at sawada@e.u-tokyo.ac.jp.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this manuscript, the author used OpenAI Codex to rephrase selected passages for language editing. After using this tool, the author carefully reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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Appendix

A News Reports on School Damage Level

Among the schools surveyed in Wenchuan County, two reported that their buildings did not collapse during the earthquake. Reports related to the Sichuan earthquake confirmed that the damage to these two schools was less severe than that to neighboring schools.
(i) The first school was Caopo Primary School. A report about survivors from Caopo Primary School published in April 2018 in *China News* (Title: One woman's memory of the Sichuan earthquake) attests that the school building did not collapse. The following excerpt is from this report:

Zhou and her younger sister were attending class at Caopo Primary School in the county that day. [Unlike thousands of other buildings in southwest China, Zhou's school somehow avoided collapse]. Zhou and all of her classmates survived the earthquake. But Zhou would soon learn that her parents and elder sister hadn't survived.

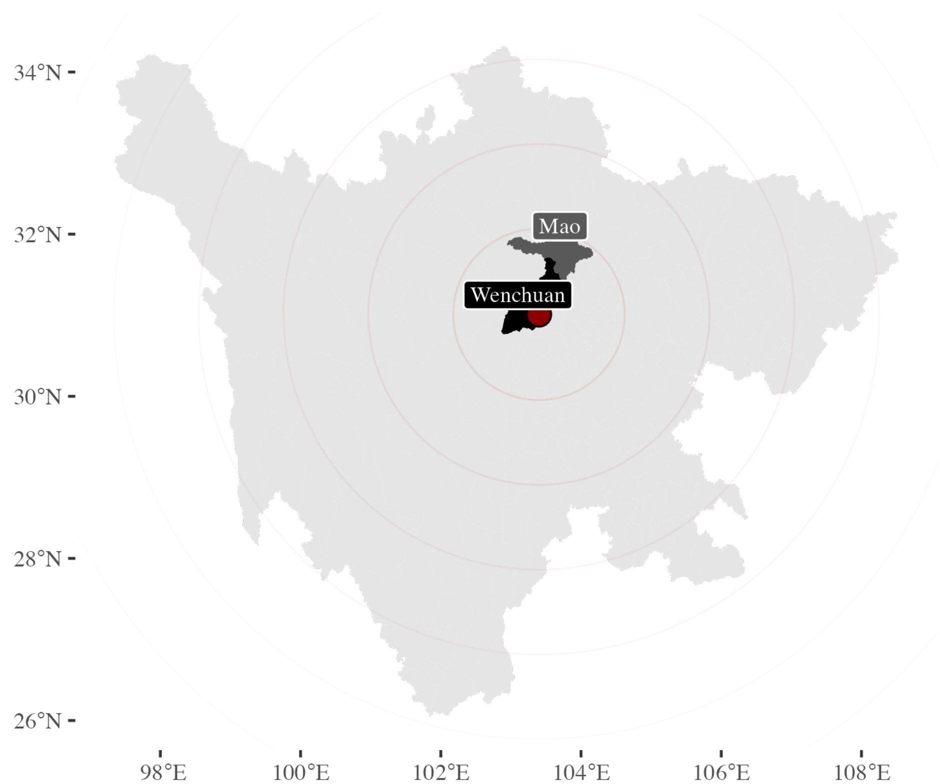
(ii) The second school is Baihua Primary School. A report in the *Zhongshan Business Daily*, in September 2008, discussing post-earthquake school reconstruction (Title: New Xuan Kou Primary School to Build the Strongest Structure) mentioned the state of Baihua Primary School. The article also noted that the school building, donated by Mr. Shao Yifu, did not likely collapse because of its high-quality construction. The following is a statement about this construction:

Upon visiting the site, it was observed that the mountainside next to the school had collapsed, and the roads were uneven. A power station near the school had three buildings, one partially collapsed and two completely. The school's gates and walls had fallen, but the main teaching building remained. An adjacent toilet block had entirely collapsed... [The teaching building, constructed in 2002 with a donation of 400,000 yuan from Mr. Shao Yifu of Hong Kong, showed some exterior wall peeling.] On closer inspection, the building bore many scars, and the steel reinforcements in many pillars were twisted and exposed. [Nevertheless, they resiliently supported the building, safeguarding the vibrant lives of hundreds of children.]

B Robustness to Omitted Variable Bias

This appendix updates the coefficient-stability exercise of Oster (2019) so that it matches the active specification used in Table 2. For each outcome, I compare column (1) and column (2)

Figure A1: Location of surveyed counties in Sichuan Province.



Notes: The map shows Sichuan Province with the two surveyed counties, Wenchuan and Mao, highlighted.

of Table 2 on the same estimation sample. Column (1) includes peer exposure, own exposure, school-grade FE, and a track indicator. Column (2) adds the relevant 2008 test score, a 2008-score missing indicator, age, gender, Tibetan minority, and class size. The logic of the test is that if the peer-exposure coefficient moves little when these observables are added, despite a sizeable increase in explained variation, then omitted variables are unlikely to overturn the main result.

Following Oster (2019), Table A1 reports two statistics. First, β^* is the implied lower-bound effect when selection on unobservables is assumed to be as strong as selection on observables ($\delta = 1$). Second, δ is the proportional degree of selection on unobservables, relative to observables, that would be required to drive the peer-exposure effect to zero. I set the attainable R-squared to 1.3 times the R-squared of Table 2 column (2) for each outcome.

Table A1: Coefficient stability of peer exposure estimates.

	Table 2 col. (2)	β^*	δ
Chinese	−4.43	−4.35	56.79
Math	−3.31	−3.02	11.31
Total	−4.01	−3.48	7.55

Notes: Each row compares columns (1) and (2) of Table 2 on the column (2) estimation sample. "Table 2 col. (2)" is the peer-exposure coefficient from the fully controlled specification. β^* is the Oster lower-bound estimate under proportional selection ($\delta = 1$). δ is the degree of selection on unobservables relative to observables required to drive the peer-exposure effect to zero. The attainable R-squared is set to 1.3 times the R-squared of Table 2 column (2).

The results point to notable coefficient stability. For Chinese and Total score, β^* remains very close to the controlled estimate in Table 2, and for Math it remains negative and similar in magnitude. The implied δ values are all positive and well above one. For each outcome, this means that selection on unobservables would need to be several times stronger than selection on the observed controls, and in the same direction, to drive the peer-exposure effect to zero. These calculations reinforce the conclusion that omitted variable bias is unlikely to account for the negative peer-exposure effects reported in Table 2.

Table A2: Summary statistics for psychosocial outcomes.

	Completion	Mean	Std. Dev.	Median
Panel A: Full Sample ($N = 1278$)				
CES-D (raw sum)	0.95	34.41	5.12	34.00
CES-D (standardized)	0.95	0.00	1.00	-0.08
SDQ (raw sum)	0.97	23.94	2.16	24.00
SDQ (standardized)	0.97	-0.00	1.00	0.03
Rosenberg (raw sum)	0.96	25.33	2.47	26.00
Rosenberg (standardized)	0.96	-0.00	1.00	-0.27
FES (raw sum)	0.96	26.57	1.75	27.00
FES (standardized)	0.96	0.00	1.00	-0.24
Panel B: Estimation Sample (Students with Non-Missing Administrative Scores) ($N = 1086$)				
CES-D (raw sum)	0.95	34.48	5.14	34.00
CES-D (standardized)	0.95	-0.00	1.00	-0.09
SDQ (raw sum)	0.98	23.95	2.15	24.00
SDQ (standardized)	0.98	-0.00	1.00	0.03
Rosenberg (raw sum)	0.97	25.32	2.49	26.00
Rosenberg (standardized)	0.97	0.00	1.00	-0.27
FES (raw sum)	0.96	26.53	1.72	27.00
FES (standardized)	0.96	0.00	1.00	-0.27

Notes: The table reports both raw-sum and standardized summary statistics for the four psychosocial scales in the survey. CES-D is the Center for Epidemiologic Studies Depression Scale; 12 of the original 20 items are available, and higher values indicate more depressive symptoms. SDQ is the Strengths and Difficulties Questionnaire; only the conduct and peer-problems subscales are observed (10 items total), and higher values indicate greater difficulties. Rosenberg is the Rosenberg Self-Esteem Scale; all 10 items are included, and the standardized measure is reverse-coded so that higher values indicate lower self-esteem. FES is the Family Environment Scale; 18 items are available, and higher values indicate more family conflict. Standardized rows are normalized to mean zero and unit standard deviation within the corresponding sample.

Table A3: Pairwise balance tests within school-grade-track cells.

Variable	Valid t-tests	Sig. at 5%
Own old-school collapse	17	0 (0.0%)
Own old-school damage scale	17	0 (0.0%)
Age	21	0 (0.0%)
Gender	21	4 (19.0%)
Tibetan minority	20	1 (5.0%)
Parental education	21	2 (9.5%)
All five characteristics	100	7 (7.0%)

Notes: All feasible classroom pairs are formed within each school-grade-track cell. The table reports the number of valid pairwise t-test comparisons and the number significant at the 5% level. "Own old-school damage scale" is the original five-point damage measure from the student survey; "Own old-school collapse" is the derived binary indicator. Tibetan minority equals 1 for students of Tibetan ethnicity. "All five characteristics" pools age, gender, Tibetan minority, parental education, and the old-school damage scale.

Table A4: Placebo regressions for predetermined student characteristics.

Outcome	Coef.	p-value	Number of observations
Age	0.76	0.20	1046
Male	−0.88	0.08	1046
Tibetan minority	−0.35	0.44	1046
Rural hukou	0.63	0.14	1033
Parental education	−0.56	0.11	1041
Pre-earthquake wealth	−2.49	0.17	1046
Household size	0.07	0.79	1046
Lives in county seat	−0.21	0.74	1043

Notes: Each row treats a predetermined student characteristic as the outcome in the whole-sample specification with peer exposure and own exposure as regressors. The table reports only the coefficient on peer exposure and its p-value. All regressions include school-grade FE and the track indicator, and are clustered at the classroom level. Reported p-values are from impose-null Rademacher wild cluster bootstrap inference with 999 classroom-level replications. Sample sizes vary because some background characteristics are missing for a subset of students.

Table A5: Effects of teaching preparation time and life-support time on academic performance.

	Chinese		Math	
	(1)	(2)	(3)	(4)
Teaching preparation	0.99		1.23	
	(0.24)		(0.40)	
Life support		−0.31		0.03
		(0.99)		(1.00)
School-grade FE	Yes	Yes	Yes	Yes
Track FE	Yes	Yes	Yes	Yes
Full controls	Yes	Yes	Yes	Yes
R2 Adj.	0.33	0.30	0.26	0.22
Number of observations	773	733	773	733

Notes: The dependent variables are standardized administrative test scores in 2010. Full controls include peer old-school exposure, own old-school exposure, the relevant 2008 test score, a missing-score indicator, age, gender, Tibetan minority, and class size. School and grade FE are included in all specifications. Wild cluster bootstrap p-values, based on 999 classroom-level Rademacher replications, are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ are based on those bootstrap p-values.

Table A6: Old-school teacher quality and school collapse.

	Old school damage
Intercept	0.94***
	(0.00)
Share of high-title teachers	−0.08
	(0.61)
Missing indicator	−0.08
	(0.55)
R2 Adj.	0.00
Number of Observations	1046

Notes: The dependent variable is an indicator for old-school collapse. The key regressor is the share of high-title teachers in the old school, with a missing-data indicator included separately. Wild cluster bootstrap p-values, based on 999 old-school-level Rademacher replications, are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ are based on those bootstrap p-values.

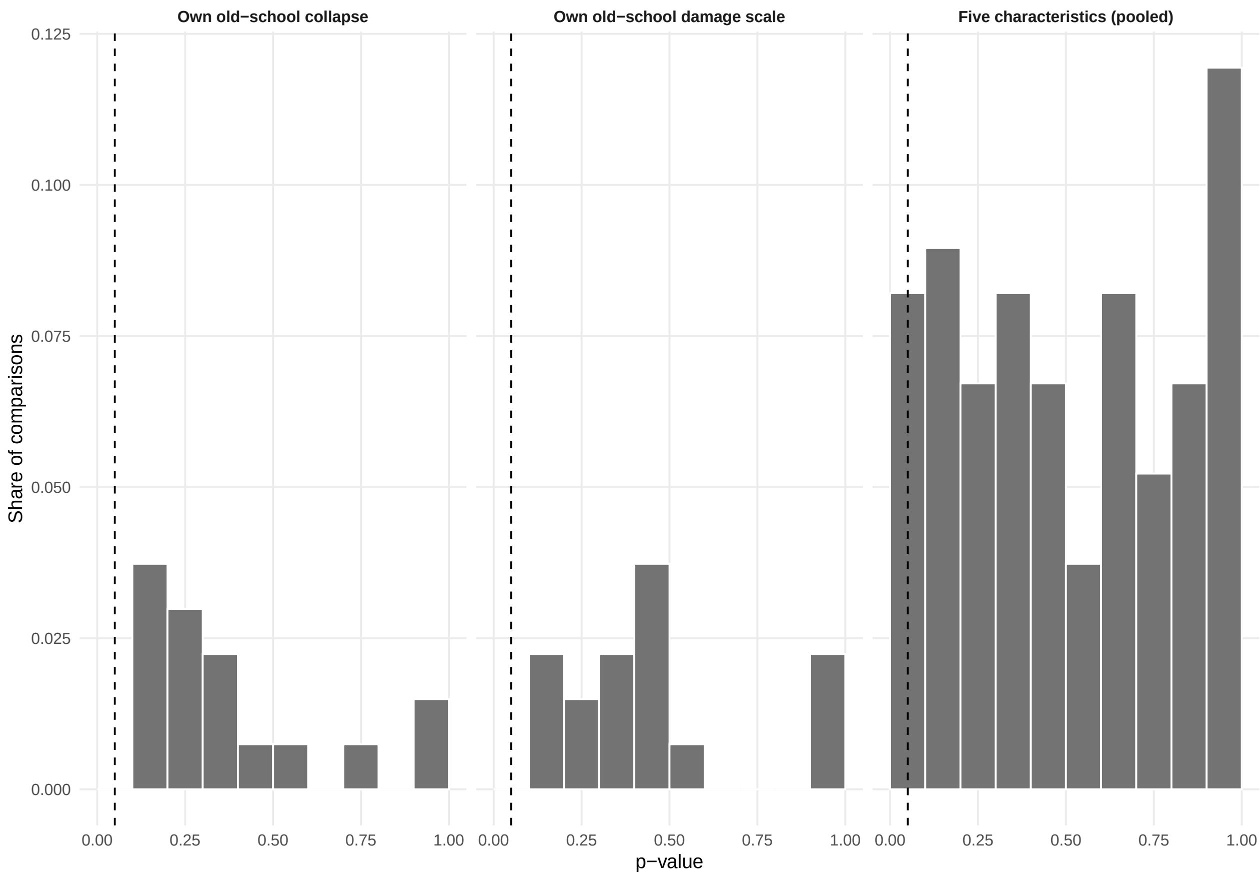


Figure A2: Distributions of pairwise-balance p-values.

Notes. The figure reports p-value histograms for pairwise t-tests within school-grade-track cells. Panels distinguish the student's own old-school collapse indicator, the corresponding five-point old-school damage scale, and the pooled set of five characteristics: age, gender, Tibetan minority, parental education, and the old-school damage scale. The collapse indicator is derived from the damage scale. The dashed line marks 0.05.

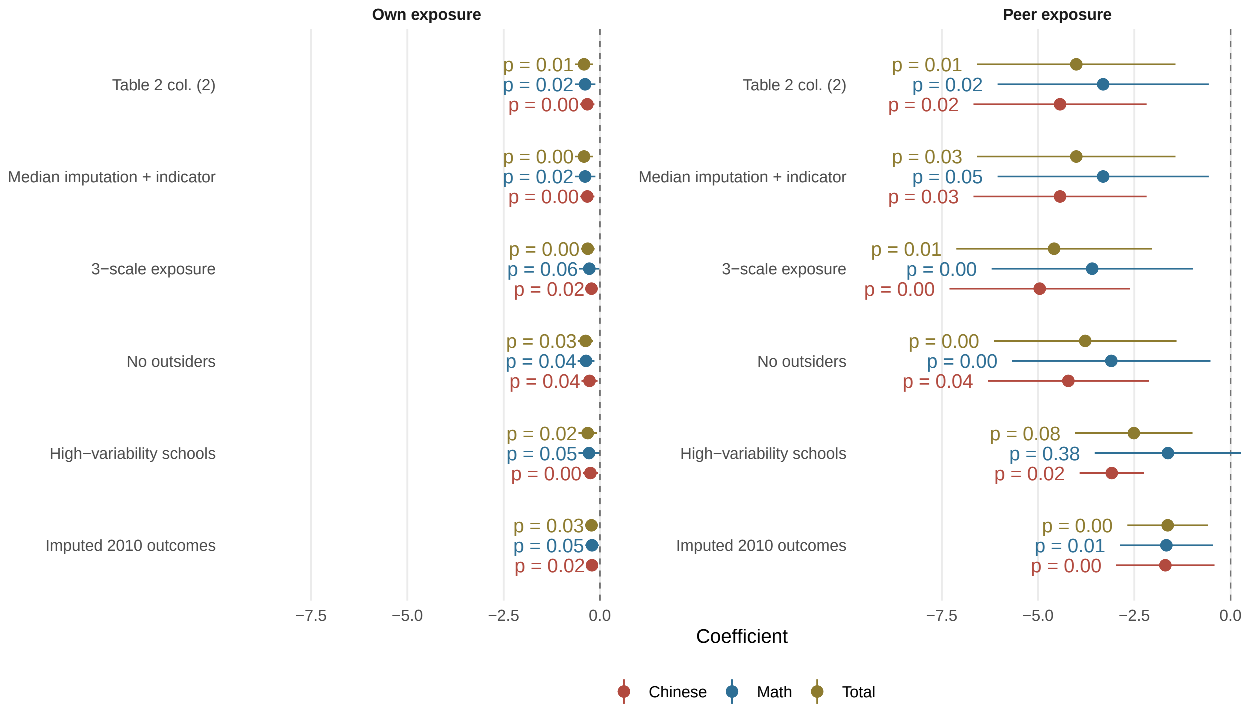
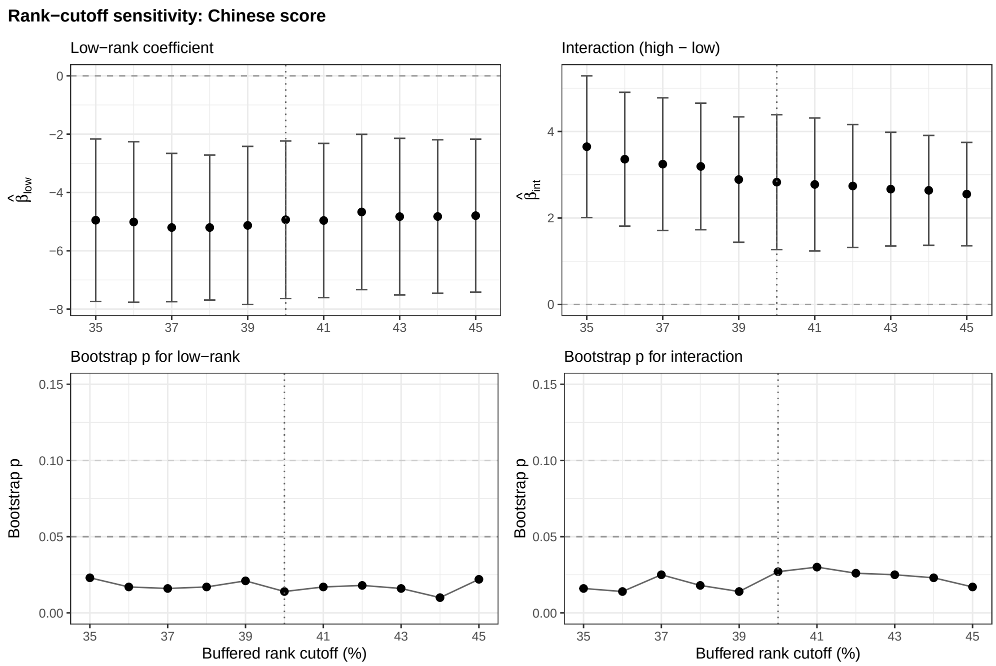
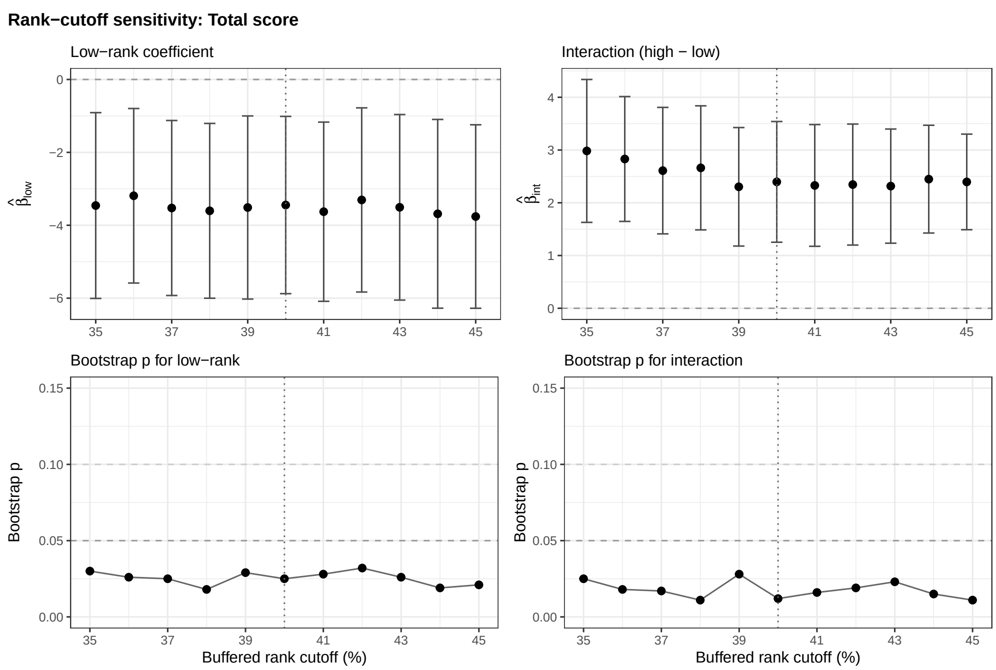


Figure A3: Robustness checks for Table 2.

Notes. Each panel reports the coefficients on own exposure and peer exposure from variants of the Table 2 column (2) specification. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. Labels to the left of the confidence intervals report p-values from wild cluster bootstrap inference. "Median imputation + indicator" replaces missing 2008 test scores with the subject-specific sample median and retains the corresponding missing-score indicator. "3-scale exposure" replaces the collapse dummy with a three-level exposure measure based on old-school damage severity. "No outsiders" restricts the sample to students whose old school is observed in the survey data. "High-variability schools" restricts the sample to schools with greater within-school variation in exposure. "Imputed 2010 outcomes" replaces missing raw 2010 administrative Chinese and math scores with the corresponding self-reported raw scores when available, then re-standardizes the combined outcomes before re-estimation. The Pearson correlation between self-reported and administrative raw scores is 0.45 for Chinese and 0.43 for Math.



(a) Chinese score



(b) Total score

Figure A4: Sensitivity of peer-exposure rank heterogeneity to the buffered rank cutoff.

Notes. Each panel varies the symmetric school-grade rank cutoff used to define low- and high-rank groups from 35 to 45 percent. Upper subpanels report the low-rank coefficient on peer exposure ($\hat{\beta}_{low}$) and the interaction coefficient ($\hat{\beta}_{int}$), with whiskers giving 95% confidence intervals from classroom-clustered standard errors. Lower subpanels report the corresponding wild cluster bootstrap *p*-values (999 replications, impose-null Rademacher). Dotted vertical lines mark the 40/40 specification used in Table 3.

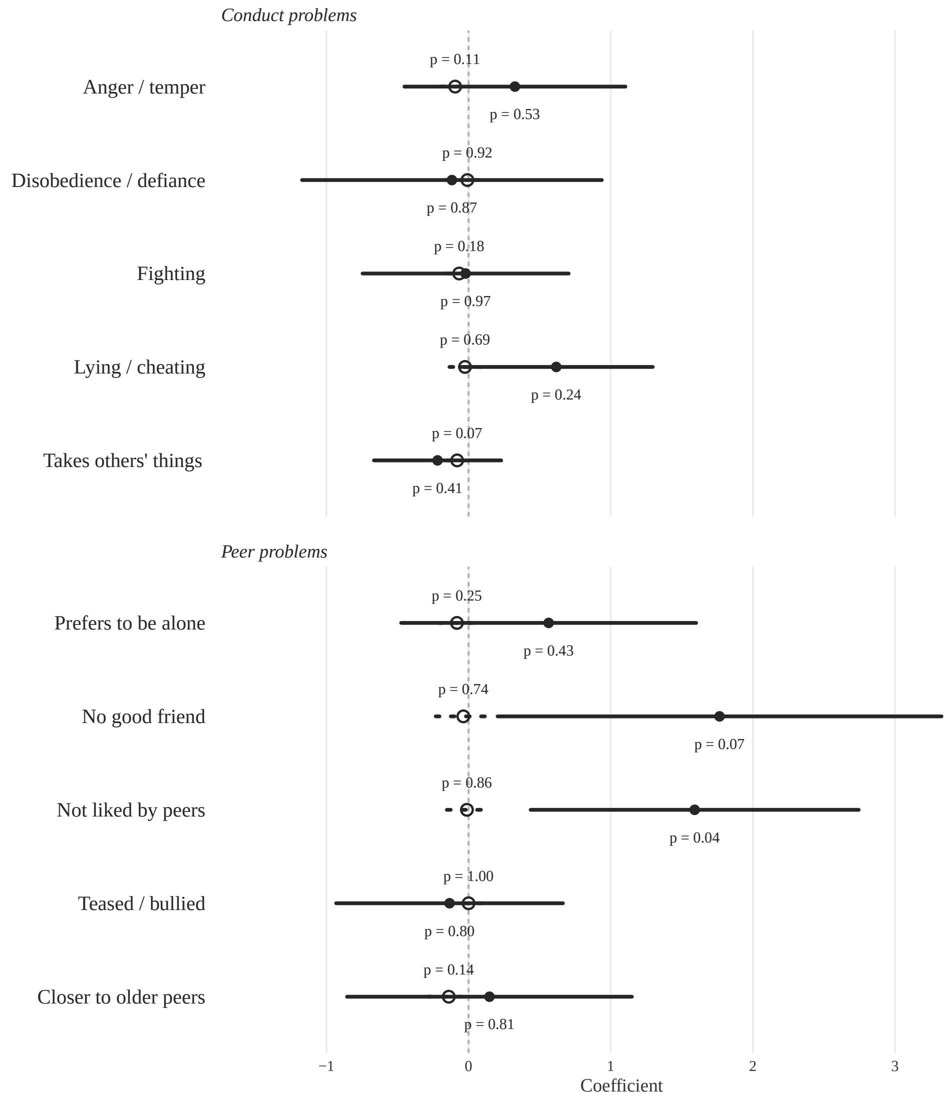


Figure A5: Item-level SDQ outcomes, by subscale.

Notes. Items are grouped by SDQ subscale: conduct problems (top) and peer problems (bottom). Filled markers (solid line) report the effect of peer exposure, measured as the share of classmates whose old school collapsed. Hollow markers (dashed line) report the effect of own exposure. Horizontal bars report 95% confidence intervals based on classroom-clustered standard errors. Marker labels report p-values based on wild cluster bootstrap inference where available and classroom-clustered p-values otherwise. Higher values indicate more behavioral or peer difficulties.

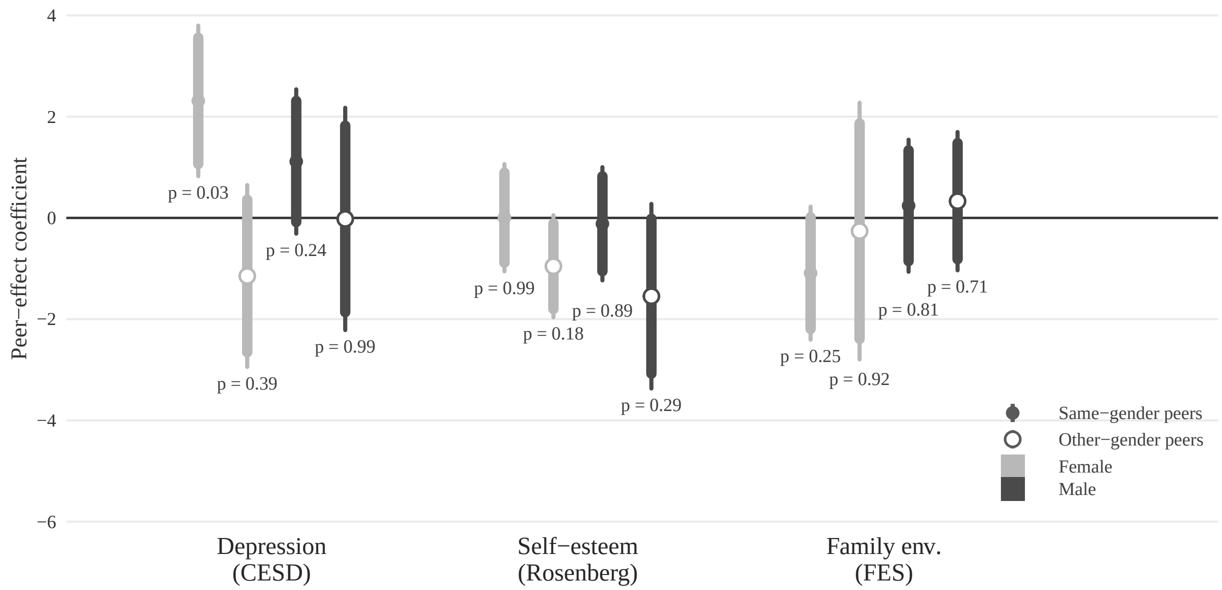


Figure A6: Same- and other-gender peer effects on psychosocial outcomes, by recipient gender.

Notes. Each panel reports peer-exposure coefficients for three psychosocial outcomes: depression (CESD), self-esteem (Rosenberg), and family environment (FES), all standardized so that higher values indicate worse outcomes. The sample is restricted to students whose own school collapsed. Same-gender peer exposure is the leave-one-out share of school-collapsed classmates of the same gender as the focal student; other-gender peer exposure is the corresponding share for classmates of the opposite gender. Coefficients are estimated separately for female and male students. Filled circles denote same-gender peer exposure; hollow circles denote other-gender peer exposure. Female estimates are shown in lighter gray and male estimates in darker gray. Wider intervals report 95% confidence intervals and narrower intervals report 90% confidence intervals, based on classroom-clustered standard errors. Labels report *p*-values from wild cluster bootstrap inference (999 replications, impose-null Rademacher). Controls are school×grade FE, track indicator, age, Tibetan minority, and class size.

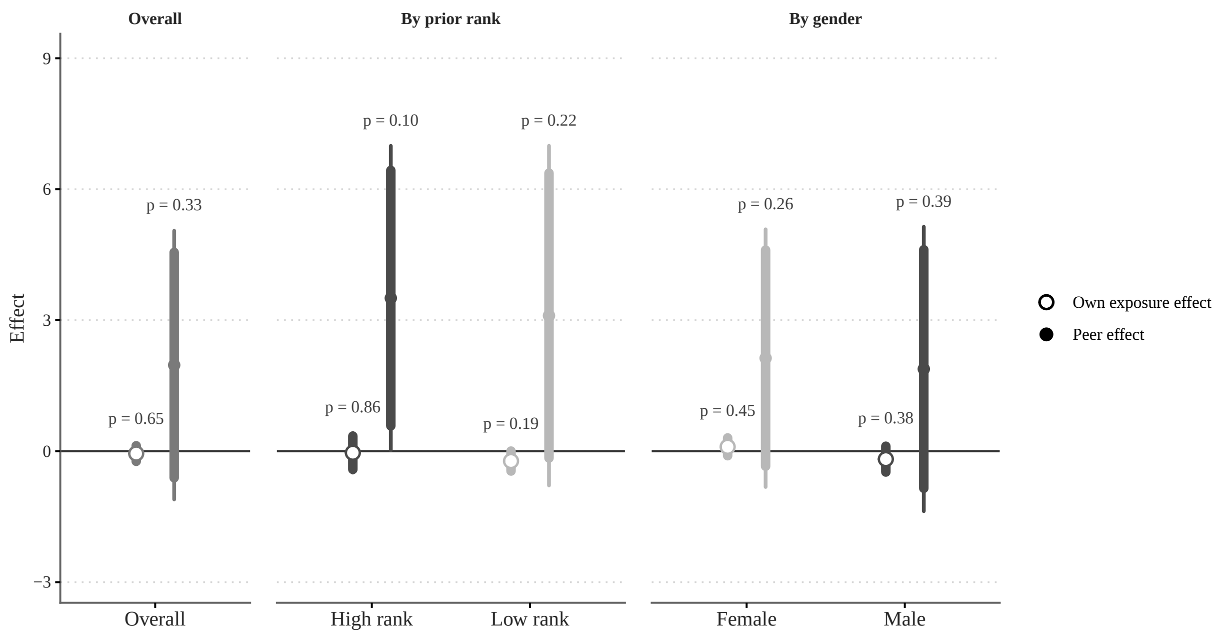
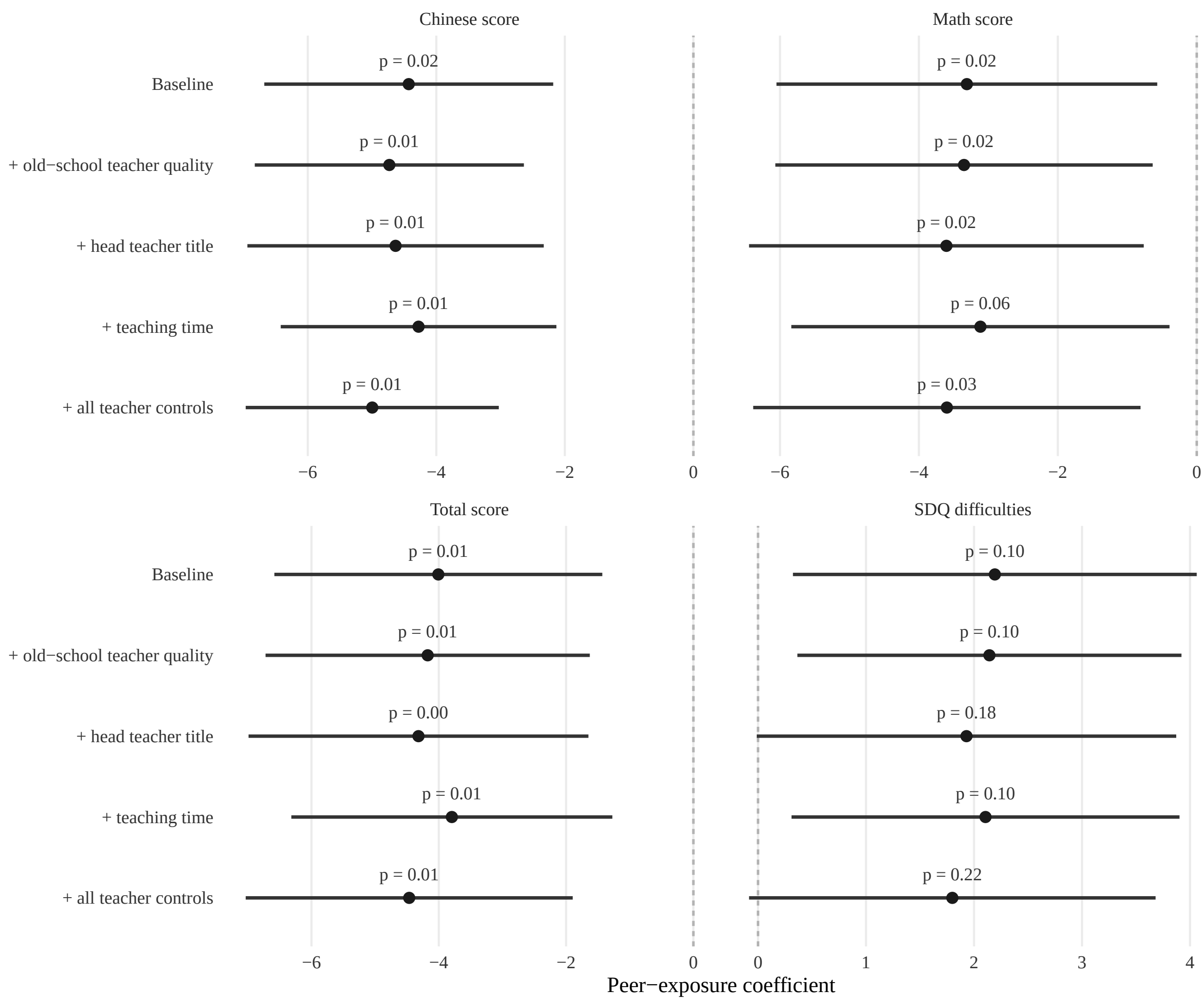


Figure A7: Study-effort estimates: overall, by prior rank, and by gender.

Notes. The figure reports study-hours estimates for three comparisons: the full sample, the prior-rank split, and the gender split. Filled circles denote peer exposure, and hollow circles denote own exposure. In the prior-rank panel, lighter gray corresponds to low-rank students and darker gray to high-rank students. In the gender panel, lighter gray corresponds to female students and darker gray to male students. Wider intervals report 95% confidence intervals and narrower intervals report 90% confidence intervals, based on classroom-clustered standard errors. Labels report p-values from wild cluster bootstrap inference (999 replications, impose-null Rademacher). All specifications include school×grade fixed effects, the track indicator, age, Tibetan minority, and class size. Own-exposure regressions additionally control for peer exposure, while peer-exposure regressions are estimated on the collapsed-school subsample. Prior rank is defined using the top and bottom 40 percent of the 2008 within-school-grade distribution.

Figure A8: Peer-exposure estimates with alternative teacher-side controls.



Notes. Each point reports the peer-exposure coefficient from the classroom-exposure specification and from versions that additionally control for old-school teacher quality, the current head teacher's title, teaching time, and all three together. Chinese, math, and total score use the same student-level specifications as Table 2. SDQ difficulties uses the collapsed-school sample. Horizontal bars show 95 percent confidence intervals based on classroom-clustered standard errors.